

Interlinks of macro-prudential policy, banking characteristics, and the lending channel of monetary transmission in Indonesia and other major Asian economies

Elis Deriantino Naiborhu¹

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Abstract

This study provides insights into determinant factors and how would interplays of these factors contribute to the credit growth in Indonesia and other major Asian economies. Results suggest that the bank lending channel exists in Indonesia and higher credit growth is attributed to prior higher liquidity level, less reliance on unstable (non-core deposits) sources of financing, higher capital buffer, lower credit risks, and lower price inefficiency. The effectiveness of accommodative macro-prudential policies to improve credit growth post-GFC 2008/09 is diminished across time. The phase I implementation (Q1 2011-Q2 2012) is effective to increase credit growth and enhances the effectiveness of lower monetary policy interest rate to boost up credit growth of large banks (the bank lending channel). However, the expected outcome has not yet been seen during the phase II of implementation (Q3 2015-Q4 2016) despite accompanied by an intense reduction of monetary policy interest rate. The ineffective impact of both loosening monetary and accommodative macro-prudential policies to improve credit growth since late 2015 is in line with the intensifying of banking risk averse behavior due to a rising credit risk since 2014 following the slowdown of economic growth that triggers them to set a higher liquidity level at the cost of lower credit growth. Further assessment confirms the policy challenges in Indonesia are to address a sensible balance of the trade-off between banking (liquidity) stability and (credit) growth and to implement measures in a timely manner.

The bank lending channel does not exist in other major Asian countries and higher credit growth is attributed to prior higher liquidity level, intense off-balance sheet activities, and less reliance on unstable (non-core deposits) sources of financing. Implementation of macro-prudential measures are effective to manage credit growth in these economies. The variance in findings of Indonesia and other major Asian countries suggests the need for detailed analysis of bank lending channel as per country case-study.

JEL: G21; G28; G32

Keywords: lending channel of monetary transmission, macro-prudential policy, banking intermediation.

Bank Indonesia and Warwick Business School. E-mail: phd15en@mail.wbs.ac. The views expressed in this paper are of the author's only and do not necessarily reflect those of Bank Indonesia and Warwick Business School. All errors are the author's only.

I. Introduction

The 2008/09 global financial crisis has taught us a valuable lesson that excessive lending without prudent behaviour of lenders in terms of overly relaxed creditworthiness requirements for borrowers would have resulted in high credit default that leads to a scarcity of cash inflow of loan repayment to lenders. The credit default has transformed into severe liquidity risk as the low rate of cash inflow does not enough to meet the high rate of cash outflow from liquidity claims of panic investors. Given the severe liquidity shortfall, banks have ceased or become very cautious in financing new investments. At the end, we have witnessed this crisis causes a deep fall in economic growth in affected countries.

The event also revealed that traditional monetary and fiscal policies failed to prevent a crisis. This fact encourages regulators particularly in Asia to implement a complementary policy of macro-prudential measures which are aimed at stabilizing the procyclicality of the financial (credit and house price) cycle thus preventing systemic risk. Following the GFC 2008/09, majority countries in Asia (China, Korea, India, Singapore, Hong Kong, Thailand, Malaysia, Philippine, Australia, and New Zealand) implement restrictive macro-prudential policies to deal with domestic's credit boom and high inflation of housing price, partly resulting from the surge of capital inflow due to accommodative monetary policies in the US and some of advanced economies in Europe (Zhang & Zoli, 2014).

While the majority of other Asian countries are dealing with the credit boom and high inflation of housing price post the GFC 2008/09, Indonesia is struggling to foster credit growth to support its economy by implementing accommodative macro-prudential policies aimed at promoting higher credit growth. Indonesia economy suffered credit crunch at the height of GFC in 2009. Credits had only grown by around 10% by the end of 2009 or only a third of the growth in 2008. After some initiatives in terms of accommodative monetary and macro-prudential policies, credit manages to grow better from 2011 until 2012 before contracting again and grows only 7.8% by the end of 2016, the lowest level in the past two decades.

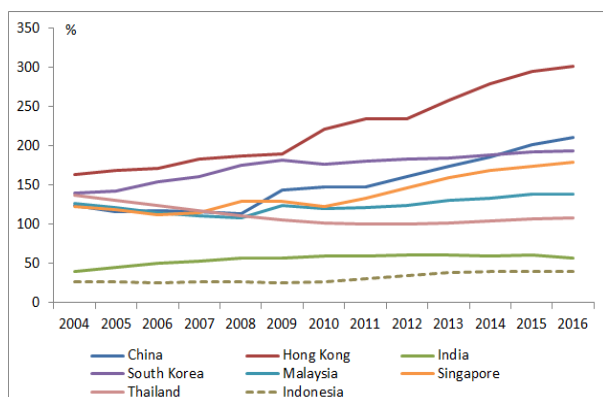
Motivation: Stylized facts of Indonesia

Fact 1: Credit growth in Indonesia

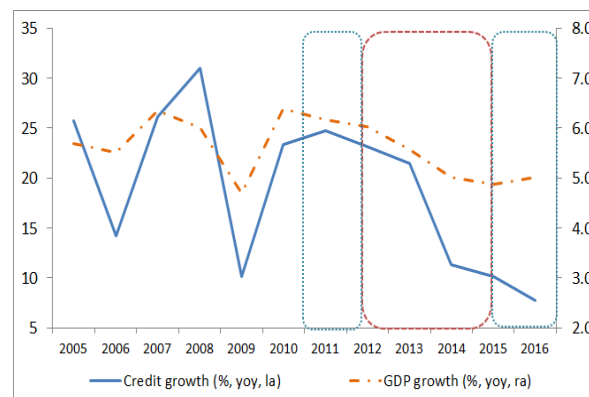
Despite the largest bank-based economy in Southeast Asia, the contribution of Indonesian banking credit to GDP is low (less than 40% by the end of 2016, which is the lowest among other major Asian economies) as indicated in Graph 1. Graph 2 shows that banking credit grows low following the 2008

GFC and both accommodative monetary policy and macro-prudential measures since late 2015. Credit grows only around 7.8% (y-o-y) by the end 2016, the lowest since the Asian financial crisis in 1997/98 where credit grew negatively.

Graph 1. Credit/GDP Major Asian Countries



Graph 2. Indonesia credit² & GDP growth



Data sources: BIS credit-to-GDP statistics, Bank Indonesia, and Financial Service Authority (OJK).

Fact 2: Policy response in managing post-crisis lending growth.

Post GFC 2008/09, Bank Indonesia (BI) adopts loosening monetary policy by reducing its monetary policy interest rate Blrate in 2009-2012 and implements an accommodative macro-prudential measure of reserve requirement linked to the loan-to-deposit ratio (RRLDR) to foster credit growth. This new regulation requires banks to maintain their intermediation activities as measured by loan-to-deposit ratio at the minimum level of 78%. To ensure banks carry out intermediation activities within manageable liquidity risk, BI sets a cap on the LDR at the level of 100%. Any bank with the intermediary level is out of the tolerable range of LDR 78%-100% will be charged a penalty in terms of a higher reserve requirement. The first phase of implementation of accommodative macro-prudential policy and loosening monetary policy is effective enough to improve credit growth. However, the high growth of credit as macro economics condition improves puts higher pressure on banking liquidity condition and leads to the housing price inflation. BI responds by initiating the implementation of the loan-to-value ratio (LTV) in 2012, continued with lowering the loan-to-value (LTV) ratio for the second, third, etc house purchasing and by lowering the upper limit of LDR to 92%. These more restrictive measures complement the tightening monetary policy of increasing Blrate since Q2 of 2013 to minimize the capital outflow due to the Fed taper issue.

Nevertheless, the combine policies that lasted until 2015 have lowered credit growth to the unprecedented level. Starting in mid-2015, BI relaxes the LTV ratio and mildly reduces the policy

² Nominal credit growth, year-on-year changes.

interest rate in mid-2015 followed by a sharp cut-off of monetary interest rate since early 2016 and increases the lower bound of LDR from 78% to 80%.

Table 1.

Date	Macro-prudential measures in Indonesia
March 2011	initial implementation of RRLDR to increase credit growth within manageable liquidity risk, tolerable LDR range:78%-100%
September 2012	initial implementation of loan-to-value ratio (LTV) with upper limit at 70%
September 2013	tightening LTV (for second, third, etc house purchases) to contain high house price growth
December 2013	decreasing upper bound of the RRLDR due to high liquidity pressure in banking system, tolerable LDR range : 78%-92%
June 2015	loosening LTV from 70% to 80%
August 2016	increasing lower bound of RRLDR to increase credit growth, tolerable LDR range: 80%-92%
	loosening LTV from 80% to 85%

Objective

Against those backdrops, I will evaluate the effectiveness of monetary interest rates pass-through and macro-prudential policy on credit growth development, the interaction of monetary interest rates policy and macro-prudential measures, and how would roles of monetary interest rate policy and macro-prudential measures on managing credit growth be altered by banking characteristics. I will focus on the role of accommodative macro-prudential measures for Indonesian credit growth. In addition to Indonesia case, I will also assess for other major Asian countries to check any similarity and variance in findings. In terms of bank characteristics, I will focus on the dimension of liquidity, equity capital, and stability source of financing of the banks as liquidity and capital in banking have been a major concern for regulators following the GFC 2008/09.

Specifically, my research will address four broad questions. (1) Do liquidity, off-b/s activities (credit commitments), the capital ratio, and the composition of funding sources affect credit growth? (2) Do liquidity, off-b/s (credit commitments) activities, the (buffer) capital ratio, and the composition of funding sources affect the pass-through of the monetary policy interest rate changes on credit growth? (3) Does implementation of macro-prudential policies affect the pass-through of the monetary interest rate changes on credit growth? (4) Does implementation of macro-prudential policies affect the role of liquidity, off-b/s activities (credit commitments), the capital ratio, and the composition of funding sources on credit growth?

A greater understanding of these issues will help regulators in identifying factors that significantly affect credit growth thus enable them to design more precise and effective policies to manage credit growth fluctuation.

Contribution

My paper will be established based on the three strands of literature review: credit growth and monetary policy (the bank lending channel), credit growth and macro-prudential policy, credit growth and banking liquidity, as well as credit growth and banking characteristics.

This research is among the first to assess the impact of the interaction of monetary policy and macro-prudential measures in affecting the credit growth or how would implementation of macro-prudential measures have affected the magnitude of banking lending channel with the case of Asian economies. The choice of Asian economies is based on the fact that these countries actively implement macro-prudential policies more than those in other territories while empirical studies on Asia case is still very limited (Zhang & Zoli, 2014). I distinguish the analysis between Indonesia and other major Asian economies as Indonesian regulators currently focus more on improving credit growth while other major Asian economies concern more on containing credit growth. Therefore, this research will provide a comprehensive assessment from two different points of regulatory concerns. The outcome is expected to enrich the literature of bank lending channel with Asian case.

This paper is structured into five parts. The second part explains relevant literature, while data, variables, hypotheses, model, some limitations, and estimation technique, are discussed in detail in part three. I present findings and robustness checks in part four. The case of other major Asia economies is presented in part five. The conclusion along with its policy implications are in part six.

II. Literature review: factors that affect banking credit growth.

Credit growth and monetary interest rates policy.

According to the bank lending channel view (Bernanke & Blinder (1992), Morris & Sellon (1995)), the volume of loans decrease and loan interest rates increase during monetary contraction, and vice versa. A tightening of monetary policy will lower banking reserves triggering a decline in bank deposits thus the availability of loanable funds. Under imperfect market capital, the decline in deposits may not be completely replaced with market-based fundings that tend to be more volatile and accessed with higher marginal cost. This tightening monetary policy then leads to the decline of banking loan supply. As tightening monetary policy increases market interest rates, the present value of fixed-rate loans will also decline, thus lower banks' net worth. In turns, this will discourage banks to supply loans resulting in lower credit growth.

Morris & Sellon (1995) also argue that institutional changes in the banking industry may alter the monetary transmission mechanism via credit channel. The growing use of loan commitments may lengthen the restrictive monetary policy transmission period because corporates can borrow under a commitment and defer the impact of a restrictive policy. The policy implication for the phenomena is the need to have an accurate judgment on the timing of policy action given high loan commitment may slow the policy transmission to constrain credit. Kashyap & Stein (1994) find that monetary policy tightening leads to a reduction in credit growth of banks in the US banks over 1976-1992. Bernanke & Blinder (1992) conclude that tightening monetary policy actually only have a small impact on lower credit growth and credit growth tends to be rigid as it takes sometimes for banks to adjust credit growth as a response to the tightening monetary policy changes (after two-quarter and lasted for around two years).

Gambacorta, *et al.*, (2011) support this view. They utilize 15 countries' data of the US and European banks over the 1999-2009 and show that tightening monetary policy leads to lower credit growth, and this effect is stronger during the crisis. They find that banks characteristics alter the transmission mechanism. Banks that rely more on core deposits change their lending less following monetary policy changes during the crisis than those rely more on short-term market instruments as a source of financing.

Credit growth and the macro-prudential policy.

The foundation theory of macro-prudential policy starts with the financial frictions in terms of limited liability and enforcement as well as asymmetric information that encourage financial agents to optimize their interests by expanding balance sheets through a higher leverage and a greater reliance on unstable short-term liquidity that one day may be the sources of negative externalities in financial intermediaries that bring market failures that in turn propagate systemic risks (Kenc, 2016). De Nicolo, *et al.*, (2012) elaborate these externalities can be in a form of strategic complementarities as banks take excessive or correlated lending strategies during the expansionary phase of financial cycle, assets fire-sales that reduce general asset prices thus impair collateral values that intensifying credit constrained during the downturn of financial cycle, and interconnectedness of systemic financial institutions through financial networks that may propagate the negative shocks during the contraction phase of the financial cycle. Macro-prudential measures are designed to correct these externalities. Specifically, the externalities related to strategic complementarities can be addressed by implementing measures that restrict the credit growth of banking. These time-varying measures need to bound banks' expected returns from taking correlated lending strategies by generating a

higher cost of having a higher credit growth during expansionary phase of the credit cycle (De Nicolo, *et al.*, 2012). Tightening measures during the upswing of credit cycles need to be loosened during the downswing of the credit cycle to reduce the fluctuation in the business cycle. Time-varying capital buffer requirement linked to credit cycle as proposed by the Basel III accord (Basel Committee, 2011), caps to loan-to-value (LTV), and debt-to-income (DTI) ratios are examples of macro-prudential measures targeted to manage banking credit growth.

Cerutti *et al.*, (2017) explain that empirical studies on the impact of macro-prudential measures on credit growth can be classified into two large groups, cross-country studies (Lim, *et al.*, (2011), Claessens, *et al.*, (2011), Zhang&Zoli, (2014), Akinci & Olmstead-Rumsey, (2015)) and micro level studies that focus on a country case with specific macro-prudential instruments such as Jimenez, *et al.*, (2012) on impact of dynamic provisioning on credit growth of Spanish banks, Igan&Kang (2011) on the implementation of loan-to-value ratio (LTV) and debt-to-income ratio (DTI) to contain mortgage growth in South Korea, Wong, *et al.*, (2011) on the effectiveness of LTV to contain the real estate cycle in Hong Kong, and Aiyer, *et al.*, (2014) on the effectiveness of a higher capital requirement on credits supply of regulated banks in UK. All these research starts from the same premise on the existence of credit boom or house price inflation problem. Both groups' assessments confirm that specific tightening macro-prudential instruments, particularly those related to housing credit are effective enough to contain credit growth or dampen procyclicality of credit and to decelerate house price growth.

Credit growth, liquidity, and off-balance sheet activities (committed loan).

Banking liquidity creation activities may expose banks to asset-liability maturity mismatch risk as banks need to provide depositors with liquidity on demand while at the same time channel illiquid credits (Diamond & Rajan, 2001). This can be mitigated by requiring banks to hold liquid assets such as cash and high-grade corporate debts and government securities (Gorton & Pennacchi, 1990) as anticipation for the uncertainty of providing liquidity to depositors. A higher ratio of liquid asset implies for less maturity mismatch risk. This liquidity condition may affect banking lending behaviour. A higher ratio of liquid assets also implies a lower level of illiquid credit. However, more liquid banks may be more confidence to expand their lending activities.

Empirical evidence, however, is still inconclusive on the impact of the liquidity ratio on credit growth. Cornett, *et al.*, (2010) find that small banks in the US with the lower ratio of liquid assets have lower credit growth. Nevertheless, Gambacorta & Marques-Ibanes (2011) find weak evidence

that a higher liquidity ratio leads to lower credit growth during the 2007-2009 crisis for the US and European banks.

In addition to collecting deposit and channeling it into credit, banking could create liquidity on an off-balance sheet via credit commitments and other liquidity calls (Kashyap *et al.*, 2002). Credit commitments are basically credit lines offered by banks to the consumers which can be drawn upon needed. The undrawn portion of these credit lines are called undisbursed committed credits and are reported as an off-balance sheet asset. When consumers draw on these lines, the off-balance sheet undisbursed credit is converted into on-balance sheet credits. Therefore, the development of undisbursed credit commitment may signal about credit demand and supply as well as potential liquidity pressure stemming from materialization of the off-balance sheet liquidity risk of which are important elements for financial system stability surveillance (Meisenzahl, 2015).

Despite its growing important role from the financial stability perspective, there are only a few empirical studies explain the relationship between undisbursed committed loan and credit growth. Cornett, *et al.*, (2010) find evidence US banks experience off-balance sheet liquidity risk as borrowers drew on the committed loan that was reflected in the higher growth of the on-balance sheet credit during the 2007-2009 crisis. Ivashina & Scharfstein (2010) also document that large draw-down of the credit lines during the financial crisis following the collapse of Lehman Brothers in 2008 contributed to the high growth of the Commercial and Investment credits in banking on-balance sheet as off-balance sheet credit commitments are converted into on-balance sheet credits. Berrospide & Meisenzahl (2015) find that liquidity shock due to tight bank funding market following the GFC in 2007/2009 has encouraged non-financial institutional firms in the US to increase the use of their credit lines to sustain their investment plans and as anticipation on limited bank's access in the future due to liquidity pressure in the banking system.

Credit growth, stability sources of funding, and equity capital.

As have been previously explained, stable funds (core deposits) can affect bank lending behaviour due to the imperfection in the capital market. The bank lending channel suggests that under an imperfection in the capital market or imperfect access to external funds, the shortfall of the bank core deposit due to tightening monetary may not be frictionless set-off by an increase of other market-based financings (wholesale funds and short-term debts) or banks may resort to funds with higher marginal costs resulting in lower credit supply (Bernanke & Blinder (1992) and Kashyap & Stein (1994)). This theory has been supported by many empirical results utilizing the US and European banking data.

Cornett, *et al.*, (2010) study the impact of liquidity risk management on credits supply in the US during the global financial crisis 2007-2009. They conclude that banks finance their lending by stable funds (core deposit and equity capital) supply more loans than the ones who rely on wholesale funds. Banks reduce their lending less if they had better access to stable deposit financing rather than those who rely more on short-term debts or unstable funds. They also find a trade-off between having less liquidity pressure and less credit supply. Gambacorta, *et al.*, (2011) find that banks with weaker capital level and rely more on unstable sources of financing and on non-interest incomes limit credit supply more during the crisis.

Another source of stable financing is equity capital. Early contributions document that capital ratio can affect bank lending behaviour. Bernanke & Lown (1991) and Peek & Rosengren (1995) empirically show that an undercapitalized position or shortage of equity capital has constrained US banks to supply loans during the 1990-1991 US recession. Under the bank lending channel of monetary policy framework, Kishan & Opiela (2000) show evidence that small and less capitalized banks in the US are most affected by the negative impact of tightening monetary policy on lower loan growth as low-capitalized banks tend to have higher agency cost that makes it more difficult to raise debt than well-capitalized banks.

In more recent studies, the relationship between credit crunch and banking capitalization also evidenced in Albertazzi and Marchetti (2010). They utilize Italian bank-firm relationship data covering 500 banks that supply credit to about 2500 non-financial firms over the period 2007-2009. They find that a shrinkage of banking loan supply is correlated with low bank capitalization as larger low-capitalized banks whose lending decision is much constrained by credit scores shifted their loan away from riskier firms following the collapse of Lehman.

Kapan & Minoiu (2013) employ syndicated loan market data of 800 banks from 55 countries over the period of 2006-2010 to explain how banks' reliance on wholesale funding and capital level may affect syndicated loan growth during the 2007/08 crisis. They show that banks that rely more on volatile market-based funding reduced the supply of syndicated loan more than other banks. However, well-capitalized banks moderate this effect as they cut their supply of loans less than other banks. This result confirms the important role of higher bank capital to the recovery of syndicated loan supply post the GFC crisis.

Studies above examine the short-run relationship between capital and loan, particularly during the downturn of economic condition. Buch & Prieto (2014) provides an explanation of the long-run relationship between capital and loan. They utilize the panel co-integration method to study the long-run impact of a higher capital level on credit growth of Germany banks over the period 1965-

2009. They find a positive long-run relationship between bank capital and loan growth as a 1% increment in the bank capital boosts credits by about 0.22%.

To sum up, all studies above consistently suggest that a higher capital level helps in stabilizing the provision of credit across economic condition, be it during crisis periods or over long term periods.

III. Methodology.

The model.

I modify the basic model of the bank lending channel of monetary transmission of Kashyap & Stein (1994) that specify real credit growth ($\Delta \log(\text{credit})$) as a function of real GDP growth ($\Delta \log(\text{GDP})$) and changes in real monetary policy interest rate (ΔCB) and extended models with bank characteristics of Gambacorta & Marques-Ibanez (2011) to take into account any structural changes in periods of implementation accommodative macro-prudential measures. I do this by adding a time dummy MPI that takes the value of 1 during the implementation of accommodative macro-prudential measures (Q1 2011-Q2 2012 and Q3 2015 - Q4 2016), the value of -1 for the tightening periods (Q3 2012-Q2 2015), and 0 otherwise, as well as its interaction terms with specific banking characteristics X and real monetary policy interest rate ΔCB . To take into account the impact of GFC 2008/09 on Indonesia banking, I also add a time dummy d_crisis that takes the value of 1 for Q3 2008-Q3 2009 and 0 otherwise. The modified bank lending channel model for bank i at time t is expressed by equation (1):

$$\begin{aligned} \Delta \log(\text{credit})_{it} = & \theta \Delta \log(\text{credit})_{t-1} + \gamma \Delta \log(\text{GDP})_{t-k} + \alpha d_crisis_t + \varphi MPI_{t-k} \\ & + (\beta + \beta^* MPI_{t-k}) \Delta CB_{t-k} + (\delta + \delta^* MPI_{t-k}) X_{it-k} + \pi X_{it-k} \Delta CB_{t-k} + \varepsilon_{it} \end{aligned} \quad (1)$$

with:

$k=1,2,3$, or 4, where k is the quarter lag. The choice of the most appropriate lag is based on its significance and the lowest score of AIC/BIC.

The implementation of accommodative macro-prudential policies in Indonesia, in general, can be grouped into two phases: phase I is Q1 2011-Q2 2012 after GFC 2008/09 and phase II since Q3 2015 following a tightening period. Therefore, I modify the time dummy MPI into two separate dummies: MPI1 and MPI2. MPI1 takes the value of 1 during the first phase of implementation (Q1 2011-Q2 2012), -1 for the tightening periods (Q3 2012-Q2 2015), and 0 otherwise, while MPI2 will be 1 during the second phase of implementation (Q3 2015 until Q4 2016), -1 for the tightening periods (Q3 2012-Q2 2015), and 0 otherwise. The modified model is presented in equation (2):

$$\begin{aligned}\Delta \log(\text{credit})_{it} = & \theta \Delta \log(\text{credit})_{t-1} + \gamma \Delta \log(\text{GDP})_{t-k} + \alpha d_{\text{crisis}_t} + \varphi_1 \text{MPI1}_{t-k} + \varphi_2 \text{MPI2}_{t-k} \\ & + (\beta_1 + \beta_1^* \text{MPI1}_{t-k}) \Delta \text{CB}_{t-k} + (\beta_2 + \beta_2^* \text{MPI2}_{t-k}) \Delta \text{CB}_{t-k} \\ & + (\delta_1 + \delta_1^* \text{MPI1}_{t-k}) X_{it-k} + (\delta_2 + \delta_2^* \text{MPI2}_{t-k}) X_{it-k} + \pi X_{it-k} \Delta \text{CB}_{t-k} + \varepsilon_{it}\end{aligned}\quad (2)$$

By comparing the coefficient of φ_1 and φ_2 , β_1^* and β_2^* , as well as δ_1^* and δ_2^* , we can trace dynamic of the effectiveness of the policy and its interaction with monetary policy as well as banking characteristics.

We can verify the first research question on the impact on credit growth in response to banking characteristics, accommodative macro-prudential measures, and changes in monetary interest rate policy by examining the statistical significance of coefficients in the δ vector, φ , and β , respectively in equation (1) and equation (2). The second hypothesis on the role of certain banking characteristics on the transmission of the bank lending channel can be tested by observing the statistical significance of interaction terms in the π vector. As an example, the distributional impact of banks liquidity on the lending channel of monetary policy, *ceteris paribus*, is expressed by:

$$\frac{\Delta \log(\text{credit})_t}{\Delta \text{CB}_{t-k}} = \beta + \pi_{LA/D} LA/D_{t-k}$$

If interaction term coefficient $\pi_{LA/D} > 0$, the credit growth of banks with higher funding liquidity in $t - k$ is less reactive to monetary policy changes, implying that a one percentage point increase in monetary interest rate policy ΔCB leads to a lower credit growth, with the decline in credit growth is less for banks with higher funding liquidity.

The third question on the role of accommodative macro-prudential measures to affect the lending channel of monetary policy can be verified by analyzing the statistical significance of the interaction term β^* . The distributional impact of accommodative macro-prudential policy on the lending channel of monetary policy can be stated as:

$$\frac{\Delta \log(\text{credit})_t}{\Delta \text{CB}_{t-k}} = \beta + \beta^* MP_{t-k}$$

If interaction term coefficient $\beta^* < 0$, the implementation of accommodative macro-prudential measures in $t - k$ causes credit growth to be more reactive to monetary policy changes. In other words, a one percentage point decrease in monetary policy ΔCB leads to a higher credit growth, with the increment growth of credit is higher than that before the implementation of macro-prudential policy.

The last question on how the role of banks characteristics to affect credit growth may be altered by the implementation of accommodative macro-prudential policy can be addressed by examining the statistical significance of the interaction terms in vector δ^* . For example, the effectiveness of

banking funding liquidity condition to increase the credit growth following the implementation of accommodative macro-prudential policy, *ceteris paribus*, is expressed by:

$$\frac{\Delta \log(\text{credit})_t}{LA/D_{t-k}} = \delta_{LA/D} + \delta_{LA/D}^*$$

If interaction term coefficient $\delta_{LA/D}^* > 0$, the implementation of accommodative macro-prudential policy in $t - k$ leads to higher credit growth, with the increment of growth is higher for banks with higher liquidity condition.

Estimation methods.

Given the dynamic nature of equation (1) and (2) due to the inclusion of lagged dependent variable of real credit growth as one of explanatory variables to capture potential persistence of credit growth, I employ Arellano-Bover/Blundell-Bond Generalized Method of Moments (GMM) system to estimate both equations. The Arellano-Bover/Blundell-Bond estimator allows the inclusion of more instruments to improve efficiency and the precision of estimator, thus offers an advantage over Arellano-Bond estimator. This method exploits additional moment conditions based on an additional assumption that first differences of instrumenting variables are uncorrelated with the fixed effects (Blundell, *et al.*, 2000).

However, Roodman (2006) mentions that a disadvantage of GMM estimator is its complication and possibility of producing bias estimates. He refers Bond (2002) and suggests for doing parameter robustness checks by comparing the parameter of lagged dependent variables estimated by GMM methods with those estimated by ordinary least square (OLS) and fixed effect least square (FELS). A stable and unbiased parameter of GMM estimators should lie within the range values of OLS and FELS parameters. This is due to the condition of the positive correlation between the lagged dependent variable and the fixed effects in the error term yields in upward biased for OLS estimator while parameters of FELS estimator tend to be downward biased due to the negative correlation between the lagged dependent variable and the error term.

Therefore, in the case GMM parameters of lagged real credit growth do not lie within the range values of OLS and FELS parameters, I will utilize the parameters estimated by FELS to do the analysis. FELS method overcomes the OLS parameter bias due to the endogeneity of lagged dependent variable to the individual effects in the error term by adding dummy of each individual bank (fixed effect). As it has been explained, FELS parameters are actually downward bias or known as dynamic Nickel bias (Bond, 2002). However, as the bias measure can be proxied with $\frac{1}{T-1}$ with T is number of

sample periods, this bias will be smaller as one employs longer time periods of observation. As I employ a relative long period of observation with $T = 48$ (quarterly data from 2005-2016), the resulting bias will be very small. Thus, parameters of FELS estimator are the most reliable ones in this case.

Therefore, to generate robust results, I will estimate equation (1) and (2) by employing Arellano-Bover/ Blundell-Bond GMM with lagged level explanatory variables as additional instrumental variables, fixed effect least square method with robust standard errors clustered at the bank level to overcome heteroscedasticity, and within-panel serial correlation problem as well as ordinary least square with robust standard errors.

Variable construction and identification issues.

The main point in the bank lending channel is that changes in the monetary policy interest rate should alter the supply of bank loans under imperfect substitutability of loans and bonds as well as rigid (sticky) prices to ensure the non-neutrality of monetary policy (Kashyap & Stein (1994) and Peek & Rosengren (1995)). Amiti & Weinstein (2013) utilize matched bank-firm dataset on credits received by listed firms in Japan to estimate the banking loan supply. As such data is not available for Indonesia, I utilize the bank-level actual credit which is the equilibrium outcome of credit supply and demand. The explanatory variables represent both demand (ΔGDP and ΔCB) and supply (ΔCB , MPI , and X) driver of credit growth. A similar approach has been applied by many previous empirical studies, including Bernanke & Blinder (1992), Gambacorta & Marques-Ibanez (2011), Zhang & Zoli (2014), and Bruno, *et al.*, (2106). Kashyap & Stein (1994) also utilize the bank-level actual credit to evaluate the bank lending channel in the US. Furthermore, they explore the cross-section difference between large and small banks and argue that supply effect may be reflected more in the credit behaviour of small banks as small banks are difficult to access external non-deposit financing due to capital market imperfection. As the consequence, credit growth of small banks will be likely more responsive to monetary policy tightening (Kashyap & Stein, 1994). Thus, distinguishing the bank lending analysis based on the banking size may alleviate the shortcoming of not utilizing credit supply growth as the dependent variable. In line with this argument, I estimate two models for all banks in the sample and then distinguish the analysis as per asset size category. A bank is categorized into the large group as per regulatory standards that sorted banks based on its asset size, business complexity, and interconnectedness with other banks or financial institutions. There are 20 large banks whose asset share is around two-third of the total industry asset while the rest of industry share belongs to 74 small banks.

The bank lending channel in Equation 1 and 2 includes banking characteristics to justify the role of supply factors in (equilibrium) credit growth while macroeconomic variable of GDP accounts for demand factors in (equilibrium) credit growth. Bank-specific characteristics X including the ratio of liquid asset to total deposit (LA/D) as a proxy for on-balance sheet funding liquidity, the ratio of undisbursed committed credit to total credit and undisbursed committed credit (committed credit) as a proxy for potential off-balance sheet liquidity pressure as well as supply and demand of credit, the ratio of non-core (unstable) funds to total deposit (stable funds) (NC/D) as a proxy of stability sources of financing, and capital buffer (*Cap_buffer*) as a proxy for banking resilience to shocks as well as a buffer of stable financing in a form of equity capital. *Cap_buffer* is defined as the deviation of a bank's realized capital adequacy ratio (CAR) to regulatory capital requirement. After hitting by the devastating impact of East Asian Financial crisis 1997/98, Bank Indonesia requires banks to maintain a minimum total CAR of 8%. As a response to the global financial crisis in 2008/09 and to strengthen banking resilience on adverse shocks, Bank Indonesia and Financial Service Authority tighten the previous capital requirement. Since 2013, each bank is required to hold a minimum CAR based on its risk profile which is reviewed twice a year. Banks are grouped into four risk profiles. Banks in lowest risk profile group or most resilience ones will be required to maintain a minimum CAR of 8%. The next risk profile group must hold a minimum CAR within the range $9\% \leq CAR < 10\%$ and the higher risk profile group must maintain a minimum CAR of $10\% \leq CAR < 11\%$. The most risky banks need to set a minimum CAR of $11\% \leq CAR \leq 14\%$.

To avoid the possible endogeneity problem between credit growth and those banking characteristics, I utilize lags of bank-specific characteristics. Following Gambacorta & Marques-Ibanez (2011), I normalize all bank-specific characteristics with regards to their average divided by standard deviation across all banks over the observation periods. This method will transform the variables into the ones that amount to zero over all observations and overcome the possibility of high multicollinearity between the explanatory variables. This method allows us to interpret the interaction parameter β and β^* as the average monetary BI interest rate impact on credit growth or credit interest rate for an average bank (Gambacorta & Marques-Ibanez, 2011).

I utilize ΔCB which is a change in the monetary real interest rate as a proxy for monetary policy stance. A positive change implies for tightening stance while a negative change indicates for loosening stance. As both macro-prudential and monetary policies take times to affect banking intermediation activities and to minimize the potential endogeneity problem from a two-way causality relationship between two policies and credit growth, I utilize lag variables of real credit growth. I also utilize a lag of quarterly changes in real GDP to account for a possible endogeneity

problem between business cycle and credit growth. Details of the explanatory variables and their hypothetical relationship with bank real credit growth are summarized in Table 2.

Variables.

Following Kashyap & Stein (1994) and Gambacorta & Marques-Ibanez (2011), the dependent variable is real credit growth $\Delta \log(\text{credit})$ which is defined as quarterly changes of lognormal of nominal banking credit to non-financial entities deflated by consumer price index.

Details of construction of explanatory variables as well as their hypothetical relationships with real credit growth are summarized in Table 3.

Table 3. Explanation on independent variables

Independent variables	Variables' description	Expected sign	Argument
Banking core variables			
LA/D	Ratio of liquid assets (cash, placement at the central bank, and high-grade securities) to deposit (%), a proxy for funding liquidity.	+	Highly liquid banks are more likely to expand credit growth (Gambacorta & Marques-Ibanez, 2011)
Committed credit	Ratio of undisbursed committed loan to sum of total asset and undisbursed committed loan (%), a proxy for potential demand and supply of loan, as well as potential off-balance sheet liquidity risk.	+/-	(+) if it serves more as potential demand and supply for loans. (-) if it indicates for higher liquidity pressure due to intense off-balance sheet activities, so that banks become more cautious to supply credit (Cornett, et al., 2010).
NC/D	Ratio of unstable funds (non-core deposit in terms of wholesale funds, short-term debts or market-based funding) to stable funds (deposit) (%), a proxy of stability sources of financing to support credit growth.	-	Banks with higher unstable funds are more subject to market volatility thus encourage banks to be more cautious to supply credit (Gambacorta & Marques-Ibanez, 2011).
Cap_buffer	Deviation of realized Capital Adequacy Ratio (CAR) to regulatory minimum capital requirement ratio (%) as a proxy	+	Banks with higher buffer are more likely to expand credit growth (Bernanke & Lown, 1991).

	for resilience to shocks and buffer of <u>stable</u> sources of funds to back up the credit supply.		
Additional banking control variables			
NPL	Ratio of non-performing loan to total loan, which is a proxy for the credit risk.	-	Higher credit risk in the past will encourage banks to put stricter requirements for borrowers leading to lower credit growth.
NIM	Net interest margin, which is a proxy for price efficiency as well as profit margin. A higher NIM could be interpreted as either higher profit or higher price inefficiency.	+/-	(+) Higher profit margin will encourage banks to extend more loans. (-) Higher price inefficiency due to higher credit interest rate and lower deposit interest rate will lead to lower demand for credit and lower demand for deposit resulting in lower credit growth.
Macroeconomic variables			
$\Delta \log(\text{GDP})$	The quarterly change of log of real GDP.	+	Expansionary of the business cycle will encourage more investment opportunities that require funding from banks.
ΔCB	The quarterly change of monetary policy interest rate adjusted with inflation rates.	-	Tightening monetary by increasing interest rate will restrict credit growth and vice versa (Kashyap & Stein, 1994, Gambacorta & Marques-Ibanez, 2011).
Macro-prudential Regulation			
MPI	Accommodative phase 1 & 2 macro-prudential policies are coded 1.	+	Policies are aimed at increasing credit growth.
MPI1	Accommodative macro-prudential policies phase 1 (Q1 2011-Q2 2012) is coded 1.	+	Policies are aimed at increasing credit growth.
MPI2	Accommodative macro-prudential policies phase 2 (Q3 2015-Q4 2016) is coded 1.	+	Policies are aimed at increasing credit growth.

The data.

I utilize quarterly data for 94 out of 117 Indonesian commercial banks over the period of 2005-2016. This sample banks cover around 90% of the Indonesian total banking asset. I exclude Syariah banks as they have different business characteristics from conventional banks. I also exclude foreign branch banks as these banks get both capital and liquidity back up from the parent company abroad. However, I include foreign subsidiary banks. Sources of banking balance sheet data and macroeconomic indicators are the Quarterly Report of Banking and Macroeconomic Statistics from Bank Indonesia and Indonesia Financial Service Authority (OJK). Individual bank data for credit risk ratio (NPL) and capitalization ratio (CAR) are extracted from the confidential banking database of Bank Indonesia and OJK.

To ensure extreme data do not affect the estimation results, I exclude observations with a capital adequacy ratio (CAR) lower than 8% as the regulator requires a bank to maintain a CAR at the minimum level of 8%. I also remove observations with CARs exceeds 95 percentile of the data. Other banking variables are trimmed should the values are less than 1 percentile and more than 99 percentile.

IV. Results

IV.1 Descriptive analysis for the Indonesia case

Table 4 displays the descriptive statistics of main variables utilized in this study for all samples, large banks, and small banks.

Table 4. Descriptive statistics

Panel A. Full sample

	$\Delta \log(\text{credit})$ (%)	LA/D (%)	NC/D (%)	Committed credit (%)	Cap_buffer (%)	NIM (%)	NPL (%)
Mean	-2.02	35.04	7.54	5.05	11.95	5.97	3.02
Median	-2.14	31.46	4.60	2.44	9.85	5.37	2.21
Min	-27.49	6.22	0.00	0.00	-2.35	1.32	0.00
Max	33.11	99.95	46.04	30.31	46.24	15.65	51.33
Std. deviation	7.41	15.73	8.35	5.89	8.00	2.59	3.83
No. obs	4419	4413	4286	4467	4226	4446	4438

Panel B. Large vs small banks

		$\Delta \log(\text{credit})$ (%)	LA/D (%)	NC/D (%)	Committed credit (%)	Cap_buffer (%)	NIM (%)	NPL (%)
Small banks	Mean	-1.97	35.13	6.26	4.75	12.46	6.15	3.03
	Std. deviation	7.70	16.56	8.02	5.75	8.50	2.59	4.08
	No. obs	3463	3453	3392	3528	3271	3490	3478
Large banks	Mean	-2.19	34.74	12.42	6.19	10.22	5.28	2.99
	Std. deviation	6.24	12.31	7.74	6.30	5.68	2.49	2.77
	No. obs	956	960	894	939	955	956	960

Table 4 Panel A shows that real credit growth (q-t-q changes, deflated by CPI) of Indonesian banks in average is very low (negative). Indonesia banks tend to be highly capitalized, as indicated by an average Cap_buffer of 11.95%. This implies that Indonesian banks have sufficient buffer to support higher credit growth. Panel B confirms that smaller banks are better capitalized than larger banks. The off –balance sheet activity in terms of undisbursed committed credit is still low, only around 5.1% of total banking asset and undisbursed committed credits with larger banks tend to be more active than smaller banks. Both large and small banks strongly depend on relatively expensive but stable deposits to finance their loans as indicated by low ratios of NC/D, 6.26% and 12.42% for small and large banks, respectively. Liquidity condition is relatively moderate with the ratio of liquid assets to deposit stands on the level around 35% for both bank groups.

Table 5. Spearman coefficient

	$\Delta \log(\text{credit})$	LA/D	NC/D	Committed credit	Cap_buffer	NIM	NPL	ΔCB	$\Delta \log(\text{GDP})$
$\Delta \log(\text{credit})$	1								
LA/D	-0.019	1							
NC/D	-0.013	-0.024	1						
Committed credit	-0.080	0.093	-0.134	1					
Cap_buffer	0.002	0.251	-0.114	0.071	1				
NIM	-0.024	-0.080	-0.018	-0.225	0.115	1			
NPL	-0.220	0.023	0.164	0.063	-0.103	0.017	1		
ΔCB	0.030	0.005	0.009	0.025	0.002	0.014	-0.005	1	
$\Delta \log(\text{GDP})$	0.118	0.162	-0.022	0.100	0.047	0.036	0.024	-0.260	1

Table 5 shows coefficients of the correlation between variables. All variables are low to moderately correlate to each other as all coefficients are less than 0.5, implying there may be no serious multicollinearity among variables.

IV.2 Regression analysis.

A. The extended lending channel of monetary transmission.

Table 7, 8a & 8b show that the estimated parameter of lagged dependent variable θ of GMM Arrelano-Bover/Blundell Bond is 0.159 that lies outside the range of FELS and OLS, which are 0.343 and 0.298, respectively. The post estimation statistics of AR(2) indicates significant values, confirming the model residual of GMM estimator suffers from second-order serial correlation problem due to inclusion of the first-differenced lagged dependent variable. These conditions imply that the estimated parameters of GMM Arrelano-Bond and Arrelano-Bover/Blundell Bond are less consistent and less robust as compared to those of FELS. In addition, the small value of θ ($\theta \ll 1$) indicates that credit growth is less persistent. Soto (2009) shows that Arrelano-Bover/Blundell Bond estimators will produce more efficient and lower bias parameters for the relatively high persistent series, or θ closes to 1. Taking into

account all the concern above, I proceed the remaining analysis based on the estimation results of FELS.

Determinant factors of credit growth.

Table 7, 8a & 8b reveal that higher funding liquidity, less reliance on non-core deposits as sources of loan financing, and higher capital buffer are key drivers of higher credit growth. The need of having higher liquidity and higher capital buffer to support higher credit growth is more evidenced among small banks.

In terms of regulation, both table 7 and 8 confirm that a positive change in ΔCB that is associated with tightening monetary policy by increasing Blrate leads to a lower credit growth, and vice versa. The economic impact of the monetary policy pass-through on credit growth is somewhat small, as 1% changes in monetary interest rate will be responded by around 0.21% changes in credit growth in the next four quarters. Both tables also indicate that a higher (lower) monetary policy interest rate is only effective to reduce (increase) credit growth of small banks and does not have significant impact on credit growth of large banks. This is inline with the theory of Kashyap & Stein (1994).

The implementation of macro-prudential measures has a moderate impact on the credit growth as indicated by MPI coefficient of 0.65. Further, Table 8a and 8b show phase I implementation of accommodative macro-prudential policy is effective to increase credit growth across all bank categories. The impact is more pronounced for large banks. It also reveals that the effectiveness of accommodative macro-prudential policy MPI to foster credit growth is diminished across time as phase II of implementation since Q3 2015 has not yet shown the expected outcome as indicated by a significant and negative coefficient of MPI2.

At last, significant and positive coefficients of $\Delta \log(GDP)$ in both table 7 and 8 confirm the pro-cyclicality of credit growth, particularly for large banks. The relatively large coefficient indicates credit growth of large banks is very much driven by demand factor of economy growth rather than the supply factors of banking characteristics. The slow down of Indonesia economic since 2012 reduces credit demand leading to a very low credit growth in the following years as verified by Graph 1.

Factors affecting the effectiveness of the monetary policy transmission

Significant and negative coefficients of interaction terms $MPI * \Delta CB$ in Table 7 indicates that implementation of macro-prudential policy enhances the effectiveness of lending channel transmission of monetary policy, particularly for large banks. However, Table 8a and 8b shows while phase I of accommodative macro-prudential implementation strengthens the monetary

policy pass-through to credit growth for large banks, the same thing does not hold for phase II implementation as indicated by non significant coefficients of interaction terms $MPI2 * \Delta CB$. This outcome helps in explaining the very low growth of credit at the end of 2016, despite the intense reduction of monetary policy interest rate since late 2015.

Certain bank characteristics enhance the effectiveness of the bank lending channel. Positive and significant coefficients of $\Delta CB * NC/D$ in Table 7 and 8a and 8b across all estimators indicates that the effectiveness of monetary interest rate policy to manage credit growth is diminished for banks with a greater reliance on unstable (non-core) sources of fundings. This applies for both large and small bank groups. Positive and significant coefficients of $\Delta CB * Committed_credit$ in Table 7 and 8 across all estimators indicates that the credit growth banks with higher undisbursed committed credit is less responsive to changes in ΔCB , particularly for small banks. This implies a higher undisbursed committed loan may serve as an indicator of a higher potential off-balance sheet liquidity pressure during tightening monetary policy while being a proxy for higher potential demand and supply of loans during loosening monetary policy. In the case of large banks, a significant and positive coefficient of $\Delta CB * Cap_buffer$ implies a higher CAR hinders the effectiveness of the lending channel transmission or the effectiveness of loosening monetary policy to increase credit growth is decreasing in capital level. Large banks with a prior higher CAR will have lower credit growth during a loosening of monetary policy but higher credit growth during a tightening of monetary policy. This indicates despite lower effectiveness of monetary policy interest rate to manage credit growth, a higher CAR reduces the fluctuation of credit growth across monetary policy phases for large banks.

Table 7. Estimation results of the bank lending channel

	Dependent Var: $\Delta \log(\text{credit})$				
	GMM Arrelano-Bover/Blundell-Bond	OLS	FELS	FELS	FELS
	All banks	All banks	All banks	Large banks	Small banks
$\Delta \log(\text{credit}) (-1)$	0.159*** [0.04]	0.343*** [0.02]	0.298*** [0.04]	0.145 [0.09]	0.325*** [0.04]
LA/D (-2)	0.548 [0.39]	0.445*** [0.14]	0.546*** [0.17]	0.155 [0.40]	0.654*** [0.19]
NC/D(-4)	-1.144*** [0.33]	-0.230** [0.11]	-0.639*** [0.20]	-1.087*** [0.36]	-0.559*** [0.21]
Committed credit (-4)	-0.750 [0.49]	-0.348*** [0.12]	-0.266 [0.17]	0.034 [0.35]	-0.327* [0.19]
Cap_buffer (-2)	0.712* [0.42]	0.541*** [0.14]	0.738*** [0.24]	-0.076 [0.34]	0.839*** [0.26]
$\Delta \text{CB} (-4)$	-0.265*** [0.08]	-0.218*** [0.07]	-0.210*** [0.06]	0.111 [0.10]	-0.300*** [0.07]
$\Delta \log(\text{GDP}) (-1)$	1.389*** [0.53]	2.047*** [0.44]	2.024*** [0.49]	3.983*** [0.90]	1.550*** [0.56]
d_crisis	-1.712*** [0.57]	-0.833* [0.44]	-0.938** [0.46]	0.325 [0.86]	-1.327** [0.54]
MPI (-1)	0.576** [0.23]	0.662*** [0.15]	0.650*** [0.14]	1.163*** [0.20]	0.587*** [0.15]
MPI (-1)* $\Delta \text{CB} (-4)$	-0.400*** [0.10]	-0.139 [0.09]	-0.131* [0.07]	-0.330*** [0.11]	-0.079 [0.08]
MPI (-1)*LA/D (-2)	-0.330 [0.41]	-0.121 [0.22]	-0.082 [0.19]	-0.273 [0.21]	-0.082 [0.23]
MPI (-1)*NC/D (-4)	0.070 [0.17]	-0.261* [0.14]	-0.310*** [0.12]	-0.246 [0.19]	-0.375*** [0.13]
MPI (-1)*Committed_credit (-4)	-0.283 [0.42]	-0.122 [0.16]	-0.127 [0.13]	0.191 [0.12]	-0.233 [0.16]
MPI (-1)*Cap_buffer(-2)	0.629 [0.40]	-0.109 [0.18]	-0.220 [0.19]	0.441* [0.27]	-0.206 [0.22]
$\Delta \text{CB} (-4)*\text{LA/D} (-2)$	-0.068 [0.08]	-0.064 [0.08]	-0.068 [0.07]	-0.069 [0.08]	-0.063 [0.08]
$\Delta \text{CB} (-4)*\text{NC/D} (-4)$	0.334*** [0.09]	0.272*** [0.07]	0.274*** [0.07]	0.222** [0.10]	0.265*** [0.08]
$\Delta \text{CB} (-4)*\text{Committed_credit} (-4)$	0.171*** [0.07]	0.167** [0.08]	0.174*** [0.07]	0.203 [0.19]	0.160** [0.06]
$\Delta \text{CB} (-4)*\text{Cap_buffer} (-2)$	0.030 [0.09]	0.124 [0.08]	0.110 [0.07]	0.248** [0.10]	0.083 [0.08]
cons	-3.001*** [0.82]	-3.709*** [0.63]	-3.735*** [0.74]	-7.157*** [1.41]	-2.946*** [0.83]
Time fixed effect	No	No	No	No	No
Bank fixed effect	No	No	Yes	Yes	Yes
R-sqr		0.20	0.18	0.16	0.19
AR 1 (z-stat)	-5.928***				
AR 2 (z-stat)	2.431**				
No obs	3411	3543	3543	773	2770

Note: Robust standard errors are reported in parentheses, with *, **, *** indicate statistical significance at the level of 10%, 5%, and 1%, respectively.

Table 8a. Estimation results of phase I and II of the MPI

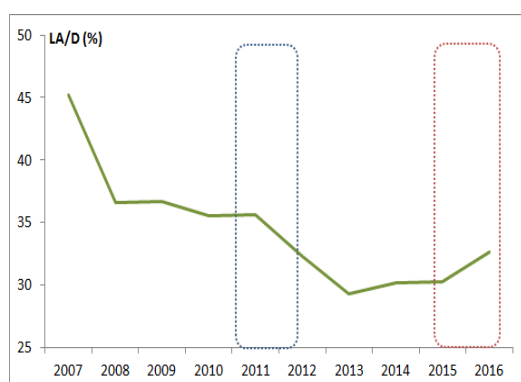
	Dependent Var: $\Delta \log(\text{credit})$				
	GMM Arrelano-Bover/Blundell-Bond	OLS	FELS	FELS	FELS
	All banks	All banks	All banks	Large banks	Small banks
$\Delta \log(\text{credit}) (-1)$	0.158*** [0.04]	0.332*** [0.02]	0.285*** [0.04]	0.132 [0.09]	0.310*** [0.04]
LA/D (-2)	0.301 [0.44]	0.323** [0.14]	0.414** [0.17]	-0.056 [0.42]	0.520*** [0.19]
NC/D(-4)	-1.110*** [0.32]	-0.225* [0.12]	-0.636*** [0.20]	-0.951** [0.35]	-0.592*** [0.21]
Committed credit (-4)	-0.729 [0.52]	-0.424*** [0.12]	-0.328* [0.18]	-0.070 [0.41]	-0.376* [0.19]
Cap_buffer (-2)	0.921** [0.47]	0.545*** [0.14]	0.780*** [0.25]	0.109 [0.38]	0.879*** [0.28]
$\Delta \text{CB} (-4)$	-0.374*** [0.08]	-0.229*** [0.07]	-0.215*** [0.06]	0.032 [0.09]	-0.287*** [0.07]
$\Delta \log(\text{GDP}) (-1)$	1.363** [0.55]	1.779*** [0.46]	1.773*** [0.48]	3.511*** [0.96]	1.361** [0.54]
d_crisis	-1.905*** [0.65]	-0.926** [0.45]	-1.036** [0.45]	-0.018 [0.92]	-1.356** [0.52]
MPI1 (-1)	0.931** [0.39]	1.337*** [0.23]	1.307*** [0.22]	2.211*** [0.40]	1.157*** [0.24]***
MPI1 (-1)* $\Delta \text{CB} (-4)$	-0.632** [0.26]	-0.141 [0.17]	-0.010 [0.15]	-0.563** [0.27]	0.012 [0.18]
MPI1 (-1)*LA/D (-2)	0.493 [1.23]	0.775** [0.31]	1.004*** [0.36]	-0.067 [0.44]	1.181*** [0.43]
MPI1 (-1)*NC/D (-4)	-0.709 [0.65]	0.137 [0.24]	0.034 [0.21]	0.146 [0.34]	-0.149 [0.21]
MPI1 (-1)*Committed_credit (-4)	-0.053 [0.84]	0.207 [0.28]	0.236 [0.20]	0.756** [0.29]	0.116 [0.25]
MPI1 (-1)*Cap_buffer (-2)	-0.475 [1.21]	-0.601** [0.26]	-0.869*** [0.31]	0.145 [0.26]	-0.970*** [0.39]
MPI2 (-1)	-0.470 [0.57]	-0.929*** [0.25]	-1.024*** [0.26]	-0.968** [0.37]	-0.988*** [0.30]
MPI2 (-1)* $\Delta \text{CB} (-4)$	-0.034 [0.15]	-0.003 [0.11]	-0.013 [0.10]	0.012 [0.16]	-0.019 [0.12]
MPI2 (-1)*LA/D (-2)	-1.160 [1.68]	-1.322*** [0.36]	-1.647*** [0.45]	-0.766 [0.59]	-1.836*** [0.52]
MPI2 (-1)*NC/D (-4)	0.703 [0.95]	-0.368* [0.22]	-0.328 [0.25]	-0.408 [0.43]	-0.251 [0.28]
MPI2 (-1)*Committed_credit (-4)	-1.500 [1.42]	-0.537** [0.27]	-0.510** [0.25]	-0.698 [0.45]	-0.501* [0.30]
MPI2 (-1)*Cap_buffer (-2)	1.261 [1.26]	0.409 [0.30]	0.444 [0.39]	0.711 [0.42]	0.556 [0.44]
$\Delta \text{CB} (-4)*\text{LA/D} (-2)$	-0.005 [0.09]	-0.038 [0.08]	-0.043 [0.07]	-0.026 [0.09]	-0.044 [0.08]
$\Delta \text{CB} (-4)*\text{NC/D} (-4)$	0.318*** [0.09]	0.270*** [0.07]	0.272*** [0.07]	0.224** [0.10]	0.265*** [0.08]
$\Delta \text{CB} (-4)*\text{Committed_credit} (-4)$	0.179** [0.08]	0.177** [0.08]	0.180*** [0.07]	0.236 [0.20]	0.159** [0.06]
$\Delta \text{CB} (-4)*\text{Cap_buffer} (-2)$	0.035 [0.09]	0.128 [0.08]	0.113 [0.07]	0.275*** [0.09]	0.089 [0.08]
cons	-2.892*** [0.88]	-3.332*** [0.67]	-3.402*** [0.72]	-6.373*** [1.47]	-2.736*** [0.81]
Time fixed effect	No	No	No	No	No
Bank fixed effect	No	No	Yes	Yes	Yes
R-sqr		0.21	0.18	0.17	0.20
AR 1 (z-stat)	-5.902***				
AR 2 (z-stat)	2.437**				
No obs	3411	3543	3543	773	2770

Note: Robust standard errors are reported in parentheses, with *, **, *** indicate statistical significance at the level of 10%, 5%, and 1%, respectively.

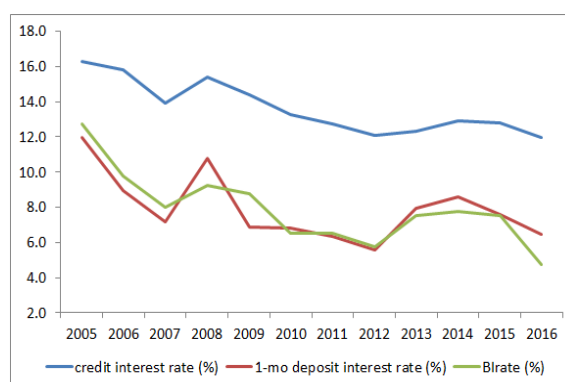
The role of bank characteristics on accommodative macro-prudential.

Table 8a confirms that the effectiveness of the phase I accommodative macro-prudential policy to improve credit growth is enhanced for banks with higher liquidity level as indicated by significant and positive coefficients of $MPI1 * LA/D$. Nevertheless, this behaviour reverses during the phase II accommodative macro-prudential policy. Banks set a higher liquidity level at the cost of lower future credit growth as confirmed by significant and negative coefficients of $MPI2 * LA/D$. Graph 3 depicts rising LA/D since 2014. This implies phase II accommodative macro-prudential poses a trade-off between a more liquid banking system and lower credit growth.

Graph 3. Development of LA/D ³



Graph 4. Retail interest rates and Blrate



Data sources: Bank Indonesia, and Financial Service Authority (OJK).

The tendency of banks to choose liquidity stability over growth during this period is also reflected in significant and negative coefficients of $MPI2 * Committed_credit$ implying that a greater committed credit in the past will put higher pressure on (off-balance sheet) liquidity of banks that in turns prohibits credit from growing higher. To strengthen the robustness role of liquidity factors on credit growth for each phase of accommodative macro-prudential implementation, I re-estimate equation (2) by including only interaction terms of both $MPI1$ and $MPI2$ with $\frac{LA}{D}$ and $Committed_credit$. The results are summarized in Table 8b that in general giving support to those of Table 8a.

Further, an analysis from price dimension of intermediation supports the current tendency of banks to choose liquidity stability over growth as shown in Graph 4. Since 2013 where credit growth decelerates, deposit interest rates tend to be higher than the Blrate, implying for a

³ The blue rectangular boxes represent phase I of accommodative periods of macro-prudential policy while the red rectangular box indicates phase II of accommodative periods of macro-prudential policy.

higher liquidity pressure in banking. In line with the increasing liquidity pressure, credit interest rate only decreases moderately and remains in a double digit despite the sharp decline in monetary policy interest rate (Blrate), thus does not provide much incentive for higher credit demand.

Table 8b. Estimation results of phase I and II of the MPI

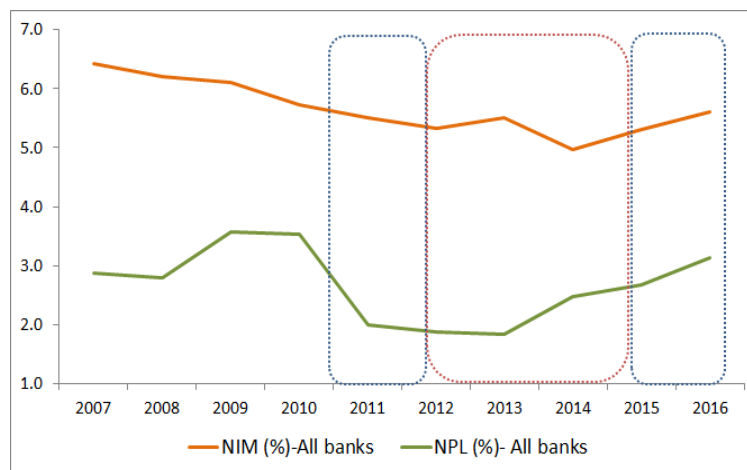
	Dependent Var: $\Delta \log(\text{credit})$				
	GMM Arrelano-Bover/Blundell-Bond	OLS	FELS	FELS	FELS
	All banks	All banks	All banks	Large banks	Small banks
$\Delta \log(\text{credit}) (-1)$	0.145*** [0.04]	0.336*** [0.02]	0.291*** [0.04]	0.133 [0.09]	0.319*** [0.04]
LA/D (-2)	0.198 [0.39]	0.324** [0.14]	0.396** [0.17]	-0.033 [0.43]	0.494** [0.19]
NC/D(-4)	-1.062*** [0.32]	-0.189 [0.11]	-0.557*** [0.20]	-0.845** [0.39]	-0.485** [0.21]
Committed credit (-4)	-0.875 [0.56]	-0.424*** [0.12]	-0.343* [0.17]	-0.115 [0.41]	-0.399** [0.19]
Cap_buffer (-2)	0.909** [0.44]	0.547*** [0.14]	0.770*** [0.24]	0.126 [0.36]	0.852*** [0.26]
$\Delta \text{CB} (-4)$	-0.368*** [0.09]	-0.217*** [0.07]	-0.203*** [0.06]	0.091 [0.14]	-0.282*** [0.07]
$\Delta \log(\text{GDP}) (-1)$	1.516*** [0.53]	1.769*** [0.46]	1.785*** [0.48]	3.443*** [0.93]	1.349** [0.54]
d_crisis	-1.773*** [0.61]	-0.976** [0.45]	-1.058** [0.45]	-0.110 [0.90]	-1.387*** [0.51]
MPI1 (-1)	0.938** [0.45]	1.322*** [0.23]	1.332*** [0.23]	2.123*** [0.47]	1.171*** [0.25]
MPI1 (-1)* $\Delta \text{CB} (-4)$	-0.712*** [0.23]	-0.184 [0.17]	-0.143 [0.16]	-0.448 [0.30]	-0.063 [0.18]
MPI1 (-1)*LA/D (-2)	0.509 [0.55]	0.542* [0.30]	0.658* [0.36]	0.033 [0.39]	0.745* [0.42]
MPI1 (-1)*Committed_credit (-4)	-0.433 [0.86]	0.104 [0.27]	0.106 [0.20]	0.656* [0.32]	-0.061 [0.23]
MPI2 (-1)	-0.438 [0.49]	-0.899*** [0.25]	-1.010*** [0.26]	-1.092** [0.42]	-0.965*** [0.30]
MPI2 (-1)* $\Delta \text{CB} (-4)$	0.033 [0.14]	0.050 [0.11]	0.041 [0.11]	-0.038 [0.19]	0.055 [0.12]
MPI2 (-1)*LA/D (-2)	-1.158 [0.70]	-1.171*** [0.34]	-1.474*** [0.41]	-0.688 [0.53]	-1.586*** [0.47]
MPI2 (-1)*Committed_credit (-4)	-0.201 [1.16]	-0.414 [0.25]	-0.383* [0.21]	-0.606 [0.47]	-0.314 [0.23]
$\Delta \text{CB} (-4)*\text{LA/D} (-2)$	0.010 [0.08]	0.009 [0.08]	0.001 [0.07]	-0.030 [0.09]	-0.001 [0.09]
$\Delta \text{CB} (-4)*\text{Committed_credit} (-4)$	0.206** [0.08]	0.171** [0.08]	0.174** [0.07]	0.170 [0.19]	0.152** [0.06]
cons	-3.210*** [0.82]	-3.300*** [0.67]	-3.393*** [0.72]	-6.233*** [1.42]	-2.675*** [0.81]
Time fixed effect	No	No	No	No	No
Bank fixed effect	No	No	Yes	Yes	Yes
R-sqr		0.20	0.17	0.15	0.19
AR 1 (z-stat)	-5.687***				
AR 2 (z-stat)	2.332**				
No obs	3499	3543	3543	773	2770

Note: Robust standard errors are reported in parentheses, with *, **, *** indicate statistical significance at the level of 10%, 5%, and 1%, respectively.

B. The extended lending channel of monetary transmission: controlling for credit risk

I include a ratio of non-performing loan to total loan (NPL) to account for variation in credit growth due to the credit risk in equation 1 and 2. Graph 3 depicts the development of credit risk (NPL), net interest margin (NIM), and capitalization ratio (CAR) of banking. The value in each year is calculated as the simple average of each respective variables for all banks in the sample. It indicates increasing trend of NPL since 2014. Rising credit risk (NPL) triggers banks to maintain high credit interest rates to have a higher profit margin to maintain a high capital buffer to absorb potential credit loss, thus contributing to low credit growth. Graph 4 confirmed this motive as it describes rising NPL since 2014 and rising margin (NIM) since 2015.

Graph 4. Development of NPL and NIM⁴



Data sources: Bank Indonesia and Financial Service Authority (OJK)

Table 9 and 10, in general, confirm the robustness of the previous conclusion of Table 7 and 8. It also indicates that a higher credit risk in the previous period significantly leads to lower credit growth of all banks. Therefore, we can infer that higher credit risks since 2014 contribute to the sharp declining of credit growth since 2014 as banks tend to be risk cautious. Moreover, significant and positive coefficients of $\Delta CB * Cap_buffer$ imply that large banks with a prior higher buffer will have lower credit growth during a loosening of monetary policy. These results provide additional explanation for the very low national credit growth during the phase II implementation of accommodative macro-prudential policy and loosening monetary stance since late 2015.

⁴ The blue rectangular boxes represent phase I of accommodative periods of macro-prudential policy while the red rectangular box indicates phase II of accommodative periods of macro-prudential policy.

Table 9. Estimation results of the bank lending channel: controlling for NPL

	Dependent Var: $\Delta \log(\text{credit})$				
	GMM Arrelano- Bover/Blundell-Bond	OLS	FELS	FELS	FELS
	All banks	All banks	All banks	Large banks	Small banks
$\Delta \log(\text{credit}) (-1)$	0.160*** [0.04]	0.333*** [0.02]	0.288*** [0.04]	0.125 [0.09]	0.317*** [0.04]
LA/D (-2)	0.636 [0.41]	0.498*** [0.14]	0.623*** [0.18]	0.446 [0.43]	0.690*** [0.20]
NC/D(-4)	-1.044*** [0.30]	-0.180 [0.11]	-0.604*** [0.21]	-1.090*** [0.38]	-0.536** [0.22]
Committed credit (-4)	-0.748 [0.46]	-0.338*** [0.12]	-0.222 [0.17]	0.087 [0.37]	-0.293 [0.19]
Cap_buffer (-2)	0.754* [0.44]	0.461*** [0.14]	0.755*** [0.24]	0.102 [0.40]	0.844*** [0.26]
NPL (-2)	-1.277** [0.65]	-0.649*** [0.15]	-0.851** [0.34]	-0.824*** [0.22]	-0.868* [0.45]
$\Delta \text{CB} (-4)$	-0.259*** [0.08]	-0.205*** [0.07]	-0.195*** [0.06]	0.123 [0.11]	-0.285*** [0.07]
$\Delta \log(\text{GDP}) (-1)$	1.211** [0.57]	2.109*** [0.44]	2.037*** [0.47]	3.837*** [0.88]	1.570*** [0.54]
d_crisis	-1.630** [0.65]	-0.826* [0.44]	-0.995** [0.45]	0.166 [0.87]	-1.364** [0.53]
MPI (-1)	0.619*** [0.22]	0.736*** [0.16]	0.732*** [0.15]	0.979*** [0.26]	0.676*** [0.16]
MPI (-1)* $\Delta \text{CB} (-4)$	-0.384*** [0.10]	-0.133 [0.09]	-0.136* [0.07]	-0.295** [0.11]	-0.087 [0.08]
MPI (-1)*LA/D (-2)	-0.270 [0.36]	-0.096 [0.22]	-0.093 [0.19]	-0.362 [0.21]	-0.080 [0.23]
MPI (-1)*NC/D (-4)	0.108 [0.17]	-0.256* [0.14]	-0.312** [0.12]	-0.156 [0.20]	-0.383*** [0.14]
MPI (-1)*Committed_credit (-4)	-0.258 [0.32]	-0.136 [0.16]	-0.139 [0.13]	0.155 [0.14]	-0.268 [0.16]
MPI (-1)*Cap_buffer (-2)	0.197 [0.31]	-0.049 [0.19]	-0.163 [0.22]	0.237 [0.23]	-0.128 [0.25]
MPI (-1)*NPL(-2)	0.001 [0.62]	-0.047 [0.24]	-0.001 [0.24]	-0.675* [0.38]	0.010 [0.25]
$\Delta \text{CB} (-4)*\text{LA/D} (-2)$	-0.076 [0.076]	-0.060 [0.08]	-0.067 [0.07]	-0.009 [0.09]	-0.060 [0.08]
$\Delta \text{CB} (-4)*\text{NC/D} (-4)$	0.297*** [0.09]	0.277*** [0.07]	0.274*** [0.07]	0.224** [0.10]	0.261*** [0.08]
$\Delta \text{CB} (-4)*\text{Committed_credit} (-4)$	0.168*** [0.07]	0.173** [0.08]	0.176*** [0.07]	0.212 [0.20]	0.162** [0.06]
$\Delta \text{CB} (-4)*\text{Cap_buffer} (-2)$	0.039 [0.08]	0.116 [0.08]	0.107 [0.07]	0.256** [0.10]	0.083 [0.08]
$\Delta \text{CB} (-4)*\text{NPL}(-2)$	0.001 [0.08]	-0.026 [0.08]	0.009 [0.07]	-0.129** [0.05]	0.061 [0.10]
cons	-2.774*** [0.89]	-3.859*** [0.63]	-3.813*** [0.73]	-6.927*** [1.36]	-3.046*** [0.82]
Time fixed effect	No	No	No	No	No
Bank fixed effect	No	No	Yes	Yes	Yes
R-sqr		0.21	0.18	0.17	0.20
AR 1 (z-stat)	-5.833***				
AR 2 (z-stat)	2.368**				
No obs	3392	3529	3529	773	2756

Note: Robust standard errors are reported in parentheses, with *, **, *** indicate statistical significance at the level of 10%, 5%, and 1%, respectively

Table 10. Estimation results of phase I and II of the MPI: controlling for NPL

	Dependent Var: $\Delta \log(\text{credit})$				
	GMM Arrelano-Bover/Blundell-Bond	OLS	FELS	FELS	FELS
	All banks	All banks	All banks	Large banks	Small banks
$\Delta \log(\text{credit})$ (-1)	0.187*** [0.05]	0.323*** [0.02]	0.276*** [0.04]	0.114 [0.09]	0.304*** [0.04]
LA/D (-2)	0.296 [0.44]	0.376*** [0.14]	0.469*** [0.18]	0.251 [0.45]	0.532*** [0.20]
NC/D(-4)	-1.058*** [0.32]	-0.181 [0.12]	-0.607*** [0.20]	-0.965** [0.36]	-0.574*** [0.22]
Committed credit (-4)	-0.562 [0.54]	-0.425*** [0.12]	-0.307* [0.18]	0.004 [0.43]	-0.369* [0.19]
Cap_buffer (-2)	0.774 [0.48]	0.473*** [0.14]	0.795*** [0.25]	0.270 [0.44]	0.889*** [0.28]
NPL (-2)	-1.692* [0.92]	-0.626*** [0.15]	-0.769** [0.31]	-0.823*** [0.22]	-0.778* [0.40]
ΔCB (-4)	-0.378*** [0.08]	-0.219*** [0.07]	-0.208*** [0.06]	0.035 [0.11]	-0.279*** [0.07]
$\Delta \log(\text{GDP})$ (-1)	1.448*** [0.56]	1.826*** [0.46]	1.801*** [0.47]	3.427*** [0.93]	1.394** [0.54]
d_crisis	-1.837** [0.77]	-0.962** [0.45]	-1.114** [0.44]	-0.147 [0.91]	-1.417*** [0.52]
MPI1 (-1)	0.829 [0.58]	1.406*** [0.24]	1.352*** [0.26]	1.953*** [0.42]	1.208*** [0.29]
MPI1 (-1)* ΔCB (-4)	-0.795*** [0.27]	-0.154 [0.16]	-0.137 [0.16]	-0.603** [0.26]	-0.032 [0.18]
MPI1 (-1)*LA/D (-2)	-0.377 [1.52]	0.695** [0.31]	0.879** [0.37]	-0.308 [0.46]	1.054** [0.44]
MPI1 (-1)*NC/D (-4)	-0.782 [0.74]	0.114 [0.24]	0.019 [0.21]	0.256 [0.34]	-0.177 [0.21]
MPI1 (-1)*Committed_credit (-4)	-0.242 [1.05]	0.248 [0.28]	0.304 [0.20]	0.786** [0.31]	0.169 [0.26]
MPI1 (-1)*Cap_buffer (-2)	-0.310 [1.24]	-0.482* [0.26]	-0.806** [0.31]	-0.051 [0.26]	-0.882** [0.34]
MPI1 (-1)*NPL(-2)	-0.289 [1.55]	0.506 [0.36]	0.466 [0.49]	-0.870 [0.70]	0.512 [0.51]
MPI2 (-1)	-0.207 [0.68]	-0.808*** [0.25]	-0.867*** [0.26]	-0.756 [0.50]	-0.834*** [0.30]
MPI2 (-1)* ΔCB (-4)	0.049 [0.17]	0.013 [0.11]	-0.003 [0.10]	0.039 [0.18]	-0.009 [0.12]
MPI2 (-1)*LA/D (-2)	-1.258 [0.39]	-1.177*** [0.36]	-1.479*** [0.45]	-0.503 [0.66]	-1.666*** [0.53]
MPI2 (-1)*Committed_credit (-4)	-0.12 [1.39]	-0.584** [0.27]	-0.578** [0.25]	-0.722 [0.47]	-0.575* [0.30]
MPI2 (-1)*Cap_buffer (-2)	1.607 [1.32]	0.370 [0.30]	0.451 [0.40]	0.821 [0.61]	0.554 [0.45]
MPI2 (-1)*NPL(-2)	0.219 [1.47]	-0.416 [0.30]	-0.305 [0.36]	0.438 [1.04]	-0.326 [0.39]
ΔCB (-4)*LA/D (-2)	-0.007 [0.09]	-0.034 [0.08]	-0.041 [0.07]	0.034 [0.09]	-0.037 [0.08]
ΔCB (-4)*NC/D (-4)	0.283*** [0.09]	0.272*** [0.07]	0.270*** [0.07]	0.230** [0.10]	0.258*** [0.08]
ΔCB (-4)*Committed_credit (-4)	0.212*** [0.08]	0.181** [0.08]	0.182*** [0.07]	0.247 [0.21]	0.161 [0.06]
ΔCB (-4)*NPL(-2)	-0.001 [0.10]	0.001 [0.08]	0.033 [0.07]	-0.127** [0.05]	0.095 [0.09]
cons	-3.006*** [0.93]	-3.424*** [0.67]	-3.460*** [0.72]	-6.251*** [1.40]	-2.810*** [0.82]
Time fixed effect	No	No	No	No	No
Bank fixed effect	No	No	Yes	Yes	Yes
R-sqr		0.21	0.19	0.18	0.20
AR 1 (z-stat)	-5.845***				
AR 2 (z-stat)	2.487**				
No obs	3392	3529	3529	773	2756

Note: Robust standard errors are reported in parentheses, with *, **, *** indicate statistical significance at the level of 10%, 5%, and 1%, respectively.

C. The extended lending channel of monetary transmission: controlling for interest margin

As a final check, I also re-run equation (1) and (2) by adding net interest margin (NIM) to control variation in credit growth due to interest margin. A higher interest margin indicates either a higher profit or a higher price inefficiency of intermediation. Graph 4 shows the tendency of higher price inefficiency since 2014 due to relatively high credit interest rates contributing to demeaned credit growth.

Table 11 and 12, in general, confirm the robustness of conclusion in previous sections. Both tables confirm that a higher price inefficiency (high NIM) leads to lower credit growth, with the impact is much more pronounced among small banks. The significant and negative coefficients of $\Delta CB * NIM$ imply that credit growth of large banks with higher price inefficiency will fluctuate more across monetary policy phases.

Table 11. Estimation results of the bank lending channel: controlling for NIM

	Dependent Var: $\Delta \log(\text{credit})$				
	GMM Arrelano-Bover/Blundell-Bond	OLS	FELS	FELS	FELS
	All banks	All banks	All banks	Large banks	Small banks
$\Delta \log(\text{credit}) (-1)$	0.153*** [0.04]	0.339*** [0.02]	0.278*** [0.04]	0.141 [0.09]	0.302*** [0.04]
LA/D (-2)	0.662* [0.39]	0.404*** [0.14]	0.598*** [0.17]	0.155 [0.39]	0.724*** [0.18]
NC/D(-4)	-1.081*** [0.33]	-0.250** [0.11]	-0.643*** [0.20]	-1.116*** [0.38]	-0.561*** [0.21]
Committed credit (-4)	-0.927** [0.46]	-0.449*** [0.13]	-0.211 [0.17]	0.106 [0.34]	-0.294 [0.18]
Cap_buffer (-2)	0.825* [0.49]	0.574*** [0.14]	0.934*** [0.24]	0.027 [0.38]	1.045*** [0.25]
NIM (-2)	-0.600 [0.43]	-0.289** [0.11]	-1.366*** [0.26]	-0.656 [0.57]	-1.418*** [0.28]
$\Delta \text{CB} (-4)$	-0.244*** [0.09]	-0.185** [0.07]	-0.177*** [0.06]	0.168* [0.09]	-0.269*** [0.07]
$\Delta \log(\text{GDP}) (-1)$	1.492*** [0.37]	2.180*** [0.45]	2.340*** [0.49]	4.155*** [0.92]	1.879*** [0.56]
d_crisis	-1.461** [0.57]	-0.686 [0.45]	-0.757* [0.44]	0.398 [0.88]	-1.127** [0.51]
MPI (-1)	0.641** [0.28]	0.720*** [0.15]	0.715*** [0.14]	1.185*** [0.19]	0.646*** [0.16]
MPI (-1)* $\Delta \text{CB} (-4)$	-0.474*** [0.13]	-0.166* [0.09]	-0.151** [0.07]	-0.359*** [0.11]	-0.102 [0.09]
MPI (-1)*LA/D (-2)	-0.156 [0.44]	-0.084 [0.23]	-0.088 [0.21]	-0.308 [0.22]	-0.075 [0.25]
MPI (-1)*NC/D (-4)	0.103 [0.19]	-0.237 [0.14]	-0.275** [0.12]	-0.283 [0.20]	-0.344** [0.14]
MPI (-1)*Committed_credit (-4)	-0.168 [0.37]	-0.051 [0.17]	-0.033 [0.16]	0.166 [0.12]	-0.138 [0.19]
MPI (-1)*Cap_buffer (-2)	0.492 [0.40]	-0.110 [0.18]	-0.222 [0.19]	0.449** [0.18]	-0.187 [0.21]
MPI (-1)*NIM(-2)	0.384 [0.49]	0.147 [0.14]	0.117 [0.18]	-0.099 [0.16]	0.151 [0.22]
$\Delta \text{CB} (-4)*\text{LA/D} (-2)$	-0.086 [0.08]	-0.081 [0.08]	-0.091 [0.07]	-0.082 [0.08]	-0.091 [0.08]
$\Delta \text{CB} (-4)*\text{NC/D} (-4)$	0.314*** [0.09]	0.262*** [0.07]	0.260*** [0.07]	0.145 [0.09]	0.261*** [0.08]
$\Delta \text{CB} (-4)*\text{Committed_credit} (-4)$	0.136** [0.06]	0.128 [0.08]	0.142** [0.07]	0.159 [0.19]	0.128* [0.07]
$\Delta \text{CB} (-4)*\text{Cap_buffer} (-2)$	0.010 [0.09]	0.150* [0.08]	0.134* [0.07]	0.291*** [0.08]	0.110 [0.08]
$\Delta \text{CB} (-4)*\text{NIM}(-2)$	-0.055 [0.07]	-0.127* [0.07]	-0.112* [0.06]	-0.219** [0.09]	-0.099 [0.08]
cons	-3.141*** [0.53]	-3.933*** [0.64]	-4.257*** [0.74]	-7.377*** [1.43]	-3.512*** [0.84]
Time fixed effect	No	No	No	No	No
Bank fixed effect	No	No	Yes	Yes	Yes
R-sqr		0.20	0.19	0.16	0.20
AR 1 (z-stat)	-5.965***				
AR 2 (z-stat)	2.474**				
No obs	3349	3504	3504	773	2731

Note: Robust standard errors are reported in parentheses, with *, **, *** indicate statistical significance at the level of 10%, 5%, and 1%, respectively.

Table 12. Estimation results of phase I and II of the MPI: controlling for NIM

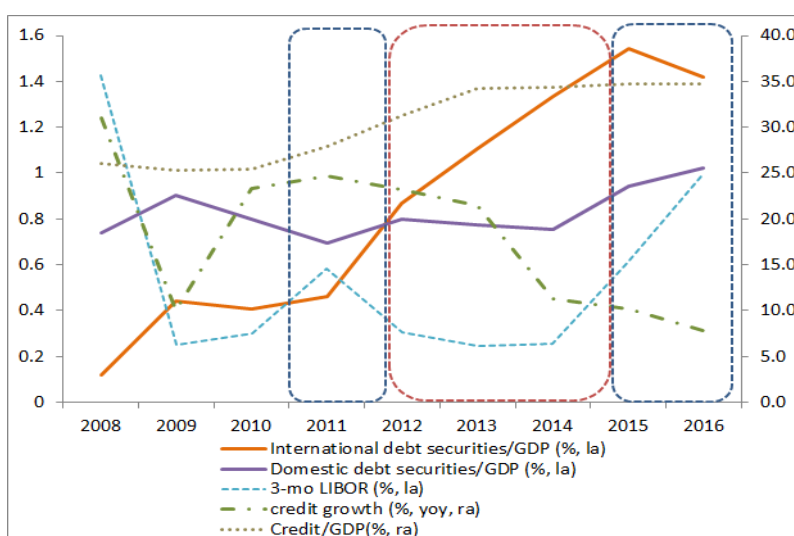
	Dependent Var: $\Delta \log(\text{credit})$				
	GMM Arrelano-Bover/Blundell-Bond	OLS	FELS	FELS	FELS
	All banks	All banks	All banks	Large banks	Small banks
$\Delta \log(\text{credit}) (-1)$	0.166*** [0.05]	0.329*** [0.02]	0.265*** [0.04]	0.129 [0.09]	0.289*** [0.04]
LA/D (-2)	0.503 [0.45]	0.292** [0.14]	0.469*** [0.17]	-0.021 [0.41]	0.586*** [0.19]
NC/D(-4)	-1.112*** [0.38]	-0.240** [0.12]	-0.626*** [0.19]	-0.978** [0.37]	-0.581*** [0.21]
Committed credit (-4)	-0.788 [0.51]	-0.526*** [0.13]	-0.271 [0.17]	-0.011 [0.40]	-0.349* [0.18]
Cap_buffer (-2)	0.748 [0.67]	0.573*** [0.14]	0.948*** [0.24]	0.152 [0.38]	1.065*** [0.26]
NIM (-2)	-0.527 [0.53]	-0.289** [0.12]	-1.322*** [0.25]	-0.532 [0.58]	-1.391*** [0.26]
$\Delta \text{CB} (-4)$	-0.372*** [0.09]	-0.194** [0.08]	-0.182*** [0.06]	0.083 [0.10]	-0.250*** [0.08]
$\Delta \log(\text{GDP}) (-1)$	1.408** [0.55]	1.853*** [0.47]	1.985*** [0.48]	3.617*** [0.97]	1.586*** [0.55]
d_crisis	-1.611** [0.68]	-0.820* [0.46]	-0.921*** [0.44]	0.018 [0.94]	-1.208** [0.51]
MPI1 (-1)	0.809 [0.54]	1.352*** [0.23]	1.360*** [0.23]	2.115*** [0.44]	1.284*** [0.25]
MPI1 (-1)* $\Delta \text{CB} (-4)$	-0.721*** [0.26]	-0.062 [0.17]	-0.024 [0.16]	-0.508* [0.28]	0.106 [0.19]
MPI1 (-1)*LA/D (-2)	0.837 [1.28]	0.795** [0.32]	0.902** [0.39]	-0.158 [0.41]	1.160** [0.48]
MPI1 (-1)*NC/D (-4)	-1.320 [1.24]	0.126 [0.23]	0.063 [0.20]	0.053 [0.34]	-0.055 [0.21]
MPI1 (-1)*Committed_credit (-4)	0.270 [1.06]	0.124 [0.29]	0.072 [0.21]	0.535 [0.37]	0.018 [0.26]
MPI1 (-1)*Cap_buffer (-2)	-0.702 [1.16]	-0.631** [0.26]	-0.843*** [0.30]	0.174 [0.24]	-0.981*** [0.39]
MPI1 (-1)*NIM(-2)	0.550 [0.70]	-0.288 [0.23]	-0.242 [0.28]	-0.415 [0.54]	-0.089 [0.33]
MPI2 (-1)	-0.649 [0.91]	-0.893*** [0.26]	-0.921*** [0.26]	-0.834** [0.40]	-1.009*** [0.30]
MPI2 (-1)* $\Delta \text{CB} (-4)$	-0.034 [0.17]	-0.061 [0.12]	-0.056 [0.11]	-0.023 [0.18]	-0.077 [0.13]
MPI2 (-1)*LA/D (-2)	-1.616 [1.82]	-1.314*** [0.37]	-1.533*** [0.45]	-0.681 [0.60]	-1.820*** [0.53]
MPI2 (-1)*NC/D (-4)	1.272 [1.41]	-0.344 [0.22]	-0.317 [0.24]	-0.351 [0.49]	-0.301 [0.27]
MPI2 (-1)*Committed_credit (-4)	-1.521 [1.52]	-0.445 [0.28]	-0.276 [0.28]	-0.515 [0.56]	-0.358 [0.31]
MPI2 (-1)*Cap_buffer (-2)	1.619 [1.20]	0.450 [0.30]	0.461 [0.38]	0.655 [0.54]	0.64 [0.42]
MPI2 (-1)*NIM(-2)	-0.341 [1.21]	0.270 [0.27]	0.27 [0.36]	0.310 [0.64]	0.074 [0.43]
$\Delta \text{CB} (-4)*\text{LA/D} (-2)$	-0.024 [0.08]	-0.062 [0.08]	-0.073 [0.07]	-0.041 [0.08]	-0.082 [0.08]
$\Delta \text{CB} (-4)*\text{NC/D} (-4)$	0.309*** [0.09]	0.260*** [0.07]	0.258*** [0.07]	0.158 [0.10]	0.260*** [0.08]
$\Delta \text{CB} (-4)*\text{Committed_credit} (-4)$	0.182* [0.10]	0.136* [0.08]	0.146** [0.07]	0.197 [0.21]	0.121* [0.07]
$\Delta \text{CB} (-4)*\text{Cap_buffer} (-2)$	0.102 [0.09]	0.154* [0.08]	0.138* [0.07]	0.313*** [0.08]	0.120 [0.08]
$\Delta \text{CB} (-4)*\text{NIM}(-2)$	-0.001 [0.08]	-0.121* [0.07]	-0.106 [0.06]	-0.191* [0.10]	-0.102 [0.08]
cons	-3.076*** [0.90]	-3.452*** [0.68]	-3.738*** [0.73]	-6.491*** [1.48]	-3.126*** [0.82]
Time fixed effect	No	No	No	No	No
Bank fixed effect	No	No	Yes	Yes	Yes
R-sqr		0.21	0.20	0.18	0.21
AR 1 (z-stat)	-5.879***				
AR 2 (z-stat)	2.506**				
No obs	3349	3504	3504	773	2731

Note: Robust standard errors are reported in parentheses, with *, **, *** indicate statistical significance at the level of 10%, 5%, and 1%, respectively.

D. A Supporting explanation: An event analysis of corporate sources of financing

To complement the role of banking as a loan provider, I do an analysis on the loan beneficiary specifically the non-financial sectors (corporations). In addition to banking loans, corporates can get financing from debt securities both in domestic and international markets. Following the GFC 2008/09 and quantitative ease (QE) policy in some developed economies leading to low international interest rates, Indonesia non-financial sectors reliance on debt securities market particularly external market increases as the cost of external debt issuance is low. As the Fed starts tapering its QE program in 2014, the cost of external debt issuance starts picking up. As a consequence, Indonesia non-financial corporates increase its utilizing of domestic debt securities market as an alternative source of financing since 2015. Increasing use of both domestic and external non-bank financing help explaining the stagnant role of banking financing contributing to very low credit growth by the end of 2016. This explanation is summarized in Graph 5 below.

Graph 5. Development of banking credit growth and Non-Financial Sector Debt Securities⁵



Data sources: Bank Indonesia, BIS debt securities statistics, FRED of Fed Res of St. Louis, and author's calculation

The policy implication of this finding is the need to develop the relatively shallow of present domestic debt securities market that may create a more competitive price of financing to banking credit interest rates for non-financial corporates.

⁵ The blue rectangular boxes represent phase I and II of accommodative periods of macro-prudential policy while the red rectangular box indicates tightening periods of macro-prudential policy.

V. The case of other major Asian economies

In this section, I will briefly discuss the case of other major Asian economies as a comparison to Indonesia case. The works of Lim, *et al.*, (2011) and Zhang & Zoli (2014) are my references to set loosening and tightening periods for Asia countries. Following Cerutti, *et al.*, (2015), I focus on the macro-prudential measures in terms of the ratio of loan-to-value, the debt-to-income ratio, the leverage ratio, capital surcharges on SIFI, limits on foreign currency credits, the reserve requirement ratio, limits on interbank exposures, credit concentration limits, limits on domestic currency credits, and the revenue levy on banking. I construct Macro-prudential Index (MPI) slightly different from the one I construct for Indonesia. As other major Asian economies focus more on restrictive macro-prudential policies to contain credit growth, for the ease of analysis I set MPI to 1, starting in the year any one of these measures was tightened until the tightening period ends and -1 if any one of these measures was lessened. However, information in Lim, *et al.*, (2011), Zhang & Zoli (2014), and Cerutti, *et al.*, (2015) are only up to 2013. Therefore, I also compile other sources, including countries FSAP reports, central bank websites, Financial Stability Reviews, and other relevant materials to update the development of macro-prudential measures in each country up to 2015. Details of periods for each phase are displayed in Appendix 1.

Data availability deserves further explanation. As I could not have complete data for all periods for all banks for the capitalization ratio CAR and committed credits, I resort to the ratio of equity to total assets and the ratio of total off-balance sheet (liabilities) items to total assets and off-balance sheet items to proxy these variables respectively. Also, I utilize changes in money market interest rates (ΔCB) as proxies for monetary policy stance with discount rates for the case of China, HongKong, and India. I describe details of the utilized variables in Appendix 2.

Moreover, as quarterly data are not fully available, I employ annual data from 2004-2005. I utilize 3 banks from Singapore, 8 banks from Hong Kong, 6 banks from Malaysia, 8 banks from Thailand, 5 banks from South Korea, 7 banks from Philippines, 16 banks from India, and 21 banks from China to make up 74 banks in total to represent major Asia countries. Most of the banks are large banks in their respective countries. As a consequence of utilizing annual data, I assign one year lag of all independent variables to avoid the possible endogeneity problem between all independent variables and credit growth. Therefore, equation (1) is rewritten as:

$$\begin{aligned} \Delta \log(\text{credit})_{it} = & \theta \Delta \log(\text{credit})_{t-1} + \gamma \Delta \log(\text{GDP})_{t-1} + \alpha d_{\text{crisis}_t} + \varphi \text{MPI}_{t-1} \\ & + (\beta + \beta^* \text{MPI}_{t-1}) \Delta \text{CB}_{t-1} + (\delta + \delta^* \text{MPI}_{t-1}) X_{it-1} + \pi X_{it-k} \Delta \text{CB}_{t-1} + \varepsilon_{it} \end{aligned} \quad (3)$$

With i is the bank index, j is the country index, and t is the year index.

To ensure outliers do not affect the estimation results, I trim banking variables at 1 and 99 percentiles.

Descriptive & regression analysis

Table 13 shows that in average other major Asian banks tend to have a lower liquid asset ratio, more off-balance sheet activities and rely more on non-core funding than Indonesian banks.

Table 13. Descriptive statistics

	$\Delta \log(\text{credit})$ (%)	NC/D (%)	offbs (%)	Eq/TA (%)	LA/D (%)	IL/TL (%)
Mean	11.22	58.19	21.31	7.75	20.96	2.98
Median	11.35	18.33	20.25	7.19	18.43	1.69
Min	-18.13	0.77	0.00	1.18	3.48	0.00
Max	66.13	1494.40	71.52	15.98	71.81	22.98
Std. dev	11.88	180.47	13.90	2.95	11.83	3.54
No.obs	784	858	844	857	858	865

Table 14 displays regression outcomes of GMM, OLS, and FELS. The estimated GMM parameter of lagged dependent variable $\Delta \log(\text{credit})_{t-1}$ is 0.199, which lies within the estimated FELS parameter of 0.175 and OLS parameter of 0.287. The post estimation statistics of AR(2) show insignificant value, implying the model residual does not have any serial correlation problem. Therefore, we can expect estimated parameters of GMM are relatively robust. I will proceed the remaining analysis utilizing GMM result.

GMM estimation shows that a higher credit growth in other major Asian countries is attributed to prior higher liquidity level, intense off-balance sheet activities, and less reliance on non-core consumer deposits as sources of loan financing. The bank lending channel does not significantly exist in these economies as the coefficient of ΔCB is not significant. This is in line with the outcome of Zhang & Zoli (2014) where interest rates does not play a significant role in determining the credit growth of 13 Asian economies over the quarterly period from 2000 to 2013 Q1. It also shows that the ineffectiveness of positive changes in monetary policy interest rate to contain credit growth is even enhanced for banks with higher liquidity and capital level as pointed by positive and significant coefficients of $\Delta CB * \frac{LA}{D}$ and $\Delta CB * \frac{Eq}{TA}$.

Tightening macro-prudential measures are effective to contain credit growth as indicated by a negative and significant coefficient of MPI. Assuming a symmetric effect of macro-prudential measures due to the MPI construction, we can infer that loosening macro-prudential measures are also effective to boost credit growth. In terms of economic impact, tightening (loosening) macro-

prudential measures moderately curb (boost) credit growth in the following year by around 3.00% as compared to an average real credit growth of 11.22%. However, banking characteristics do not have a significant impact on altering the effectiveness of macro-prudential measures on credit growth.

Table 14. Estimation result of the bank lending channel in Asian economies

	Dependent Var: $\Delta \log(\text{credit})$					
	GMM Arrelano-Bover/Blundell-Bond	OLS	FELS	GMM Arrelano-Bover/Blundell-Bond	OLS	FELS
	All banks	All banks	All banks	All banks	All banks	All banks
$\Delta \log(\text{credit}) (-1)$	0.199** [0.09]	0.287*** [0.04]	0.175*** [0.04]	0.169* [0.09]	0.264*** [0.04]	0.150*** [0.04]
$LA/D (-1)$	4.911*** [1.83]	0.756 [0.599]	1.568* [0.82]	4.229** [1.89]	0.612 [0.59]	1.280 [0.79]
$NC/D(-1)$	-5.878*** [2.16]	-0.265 [0.33]	-4.207** [1.61]	-5.665** [2.21]	-0.241 [0.32]	-3.951** [1.64]
$Offbs (-1)$	3.286** [1.63]	1.584*** [0.54]	2.853*** [0.70]	3.539** [1.60]	1.581*** [0.54]	2.812*** [0.70]
$Eq/TA (-1)$	-2.130 [2.08]	-0.069 [0.65]	-2.722* [1.55]	-2.155 [2.08]	0.017 [0.64]	-2.599* [1.50]
$\Delta CB (-1)$	-0.093 [0.28]	0.125 [0.19]	0.083 [0.18]	0.124 [0.25]	0.250 [0.18]	0.199 [0.17]
$\Delta \log(GDP) (-1)$	0.613* [0.36]	1.134*** [0.18]	0.923*** [0.31]	0.654* [0.39]	1.165*** [0.18]	0.955*** [0.31]
d_crisis	-4.048** [2.00]	-2.288* [1.29]	-2.072* [1.22]			
$MPI (-1)$	-3.000*** [0.76]	-2.876*** [0.52]	-2.504*** [0.50]	-2.700*** [0.80]	-2.644*** [0.51]	-2.285*** [0.50]
$MPI (-1)*\Delta CB (-1)$	0.278 [0.26]	0.090 [0.17]	-0.021 [0.14]	0.132 [0.27]	-0.003 [0.16]	-0.108 [0.15]
$MPI (-1)*LA/D (-1)$	-0.273 [1.26]	0.902 [0.60]	0.201 [0.55]	0.065 [1.29]	1.036* [0.59]	0.327 [0.56]
$MPI (-1)*NC/D (-1)$	0.036 [0.81]	-0.182 [0.34]	-0.064 [0.32]	0.113 [0.76]	-0.145 [0.35]	-0.039 [0.33]
$MPI (-1)*Offbs (-1)$	-0.562 [0.84]	-0.036 [0.55]	-0.341 [0.49]	-0.860 [0.80]	-0.109 [0.55]	-0.433 [0.48]
$MPI (-1)*Eq/TA(-1)$	-1.358 [1.28]	-0.445 [0.63]	-0.484 [0.76]	-1.015 [1.27]	-0.402 [0.62]	-0.448 [0.76]
$\Delta CB (-4)*LA/D (-1)$	0.979*** [0.30]	0.433** [0.19]	0.406** [0.18]	0.966*** [0.31]	0.452** [0.19]	0.425** [0.18]
$\Delta CB (-4)*NC/D (-1)$	-0.328 [0.27]	-0.147 [0.10]	-0.172 [0.11]	-0.318 [0.23]	-0.142 [0.10]	-0.170 [0.12]
$\Delta CB (-4)*Offbs (-1)$	0.119 [0.30]	-0.044 [0.20]	-0.031 [0.15]	0.061 [0.30]	-0.062 [0.20]	-0.042 [0.15]
$\Delta CB (-4)*Eq/TA (-1)$	1.044*** [0.38]	0.769*** [0.20]	0.758*** [0.19]	1.025*** [0.37]	0.778*** [0.20]	0.764*** [0.19]
cons	9.671*** [2.05]	5.127*** [1.10]	8.570*** [1.89]	9.109*** [2.18]	4.800*** [1.10]	8.291*** [1.86]
Time fixed effect	No	No	No	No	No	No
Bank fixed effect	No	No	No	No	Yes	Yes
R-sqr		0.31	0.19		0.31	0.18
AR 1 (z-stat)	-4.967***			-4.871***		
AR 2 (z-stat)	0.897			0.242		
No obs	517	653	653	517	653	653

Note: Robust standard errors are reported in parentheses, with *, **, *** indicate statistical significance at the level of 10%, 5%, and 1%, respectively.

Table 14 also reveals that demand factor of GDP has a moderate impact on credit growth implying that the procyclicality of credit growth in these economies is relatively moderate. In these economies, supply factors play a greater role in determining credit growth. At last, as the coefficient of d_crisis is relatively large than those estimated by OLS and FELS, I re-run the regression by excluding d_crisis to ensure the conclusion is not driven by d_crisis . Results in Table 14, in general, confirm the robustness of conclusion.

Limitation of the study

A robustness check for the presence of the bank lending channel in other Asian economies can be done by utilizing higher frequency (quarterly) data. This is particularly important because there is a possibility that credit growth's response to a change in monetary interest rate policy is earlier than a year. Kashyap & Stein (1994) shows that credit growth of banks in US response to monetary policy interest rate two quarters after the shock and take around two years for the whole transmission. However, the limited availability of public individual bank data is my greatest constraint of doing analysis utilizing quarterly data.

Another limitation of this Asian cross country study merits discussion. Kashyap & Stein (1994) argue the bank lending channel behaviour of large and small banks are different due to the different ease of access to external non-deposit financing and show for the US case that the credit growth of small banks is more sensitive to monetary interest rate changes than that of large banks. The same thing holds for Indonesia case. Due to the data constraint, assessment on other major Asian economies in this study utilizes mostly data of large banks. There is a possibility that the bank lending channel may exist among small banks in these economies. Therefore, the assessment of the bank lending channel needs to distinguish between large and small banks, particularly in the case it is difficult to distinguish between loan supply and demand due to data constraint. It is also important to utilize a more specific data set to verify the earlier indication that supply factors play a greater role than demand factors in determining credit growth in these economies.

VI. Conclusion and Policy implication

The bank lending channel exists in Indonesia and higher credit growth is attributed to higher liquidity level, less reliance on unstable (non-core deposits) sources of financing, stronger capital buffer, lower credit risks, and lower price inefficiency. There are some distinguishing features between bank categories. The credit growth of small banks tends to be more sensitive to funding liquidity condition, the resilience of capital buffer, price inefficiency, and monetary policy interest rate

changes while credit growth of large banks is much more affected by reliance on stable (core deposits) sources of financing, demand factor of economic growth as well as implementation of macro-prudential policy.

Bank characteristics also affect the effectiveness of bank lending channel transmission. A greater reliance on unstable (non-core deposits) sources of financing and a higher off-balance sheet of committed credits reduce the effectiveness of lower monetary policy interest rate to foster credit growth. Regulators may consider implementing the net stable funding ratio (NSFR) to reduce reliance on unstable sources of fund, particularly for large banks. A higher capital buffer reduces the fluctuation of credit growth, thus ensuring the stability of credit growth of large banks across monetary policy phases. Meanwhile, a greater price inefficiency less stabilizing the credit growth of large banks across monetary policy phases.

The effectiveness of accommodative macro-prudential measures to promote higher credit growth is diminished across times. The phase I implementation has a moderate impact on increasing credit growth and enhances the effectiveness of loosening monetary interest rate policy to foster credit growth among large banks, while the expected outcome has not yet been seen during the implementation of phase II.

The ineffective impact of both loosening monetary and accommodative macro-prudential policies to improve credit growth adopted since late 2015 are attributed to four factors. First, low credit growth in Indonesia is more on demand constraint rather than supply (banking characteristics) factors. This particularly applies to the case of large banks. The slowdown of Indonesia economic contributes much to the very low credit growth by the end of 2016. Second, the analysis from loan beneficiaries (non-financial corporates) shows an increasing utilizing of non-bank financing both domestic and external debt securities markets. The implication policy of this finding is the need to develop the relatively shallow of present domestic debt securities market in order to create a more competitive price of financing to banking credit interest rates for non-financial corporates. Third, deteriorating credit quality in terms of increasing non-performing loans since 2014 following the slowdown in economic growth triggers banks to maintain high credit interest rates to have a higher interest margin to absorb credit loss. This intensifying of risk cautious behaviour encourage banks to set a higher liquidity level at the cost of lower credit growth, implying for a trade-off between a greater banking liquidity stability and lower credit growth. Fourth, timing and type of accommodative measures taken by the regulator. Despite a sharp reduction of credit growth in 2014, the regulator keeps imposing tightening policies to address higher liquidity and credit risk pressure until Q2 2015 implying a choice of banking stability over growth. It was only in Q3 2015 Bank Indonesia started to

implement phase II accommodative macro-prudential policy by increasing upper limit of LTV from 70% to 80%. As mortgage contributes only around 12% of total banking credit, the impact of increasing the LTV ratio from 70% to 80% on overall credit growth is not significant. In addition, the monetary policy rate is only reduced for 25 bps in 2015. Significant efforts in pursuing loosening policies were actually started in Q3 2016 by slashing monetary interest rate for 275 bps and by increasing the lower limit of RRLDR from 78% to 80% as well as the LTV ratio from 80% to 85%. As both policies need times to be transmitted, the expected improvement in credit growth has not yet been seen by the end of 2016.

The findings above confirms the policy challenges in Indonesia are to address a sensible balance of the trade-off between banking (liquidity) stability and (credit) growth and implement measures in a timely manner.

The bank lending channel does not exist in other major Asian countries and higher credit growth is attributed to higher liquidity level, intense off-balance sheet activities, and less reliance on unstable (non-core deposits) sources of financing. Implementation of macro-prudential measures has a moderate impact on managing credit growth in these economies. There is an indication that supply factors play a greater role than demand factors in determining credit growth in these economies. The variance in findings of Indonesia and other major Asian countries suggests the need for detailed analysis of bank lending channel as per country case-study utilizing a detailed data set, due to high heterogeneity in the nature of policies and banking industry profile of each country.

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Appendix 1. Implementation phases of macro-prudential measures in major Asia countries

	Loosening periods	Tightening periods
Singapore	2004-2009	2010-2015
HongKong	2004-2008	2009-2015
Malaysia	2008-2009	2004-2007, 2010-2015
China	2008-2009, 2014-2015	2004-2007, 2010-2013
India	2009	2004-2008, 2010-2015
South Korea	2008, 2010, 2014-2015	2004-2007, 2009, 2011-2013
Thailand	2008-2009	2004-2007, 2010-2015
Philippine	2008-2010	2004-2007, 2011-2015

Appendix 2. Variables and data sources of major Asia countries

Variable	Definition	Data source
$\Delta \log$ (credit)	The annual change of log of real gross loan	Bankscope/Orbis bank focus
LA/D	Ratio of liquid assets to deposits and short term funding	Bankscope/Orbis bank focus
Offbs	Ratio of off- balance sheet items to total asset and off-balance sheet items. This is a proxy for (undisbursed) committed credit given data series of committed credit is not available for entire periods.	Bankscope/Orbis bank focus
NC/D	Ratio of non core funding to consumer deposits. Non core funding is calculated as total funding-consumer deposits	Bankscope/Orbis bank focus
EQ/TA	Ratio of total equity to total asset. This is a proxy for capitalization level given capital adequacy ratio is not available for all banks in all periods.	Bankscope/Orbis bank focus
ΔCB	Central bank policy interest rate. It is proxied by real discount rates for the case of China, HongKong, and India and real money market interest rate for other countries.	International Financial Statistics, IMF
$\Delta \log$ (GDP)	The annual change of log of real GDP	Federal Reserve Economic Data, Federal Reserve Bank of St. Louis
MPI	Macro prudential index. Code 1 indicates tightening periods and code -1 for loosening periods.	Zhang & Zoli (2014), Lim, <i>et al.</i> (2011), and Author's calculation